LSG 770-154-10 - LM-5

# LUNAR MODULE

# STRUCTURES

# HANDOUT

# LM-5

FOR TRAINING PURPOSES ONLY; IN
SUPPORT OF LM-5 STRUCTURES COURSE

FOR

FLIGHT CONTROL DIVISION

AT

MSC, HOUSTON, TEXAS


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MAY 1969

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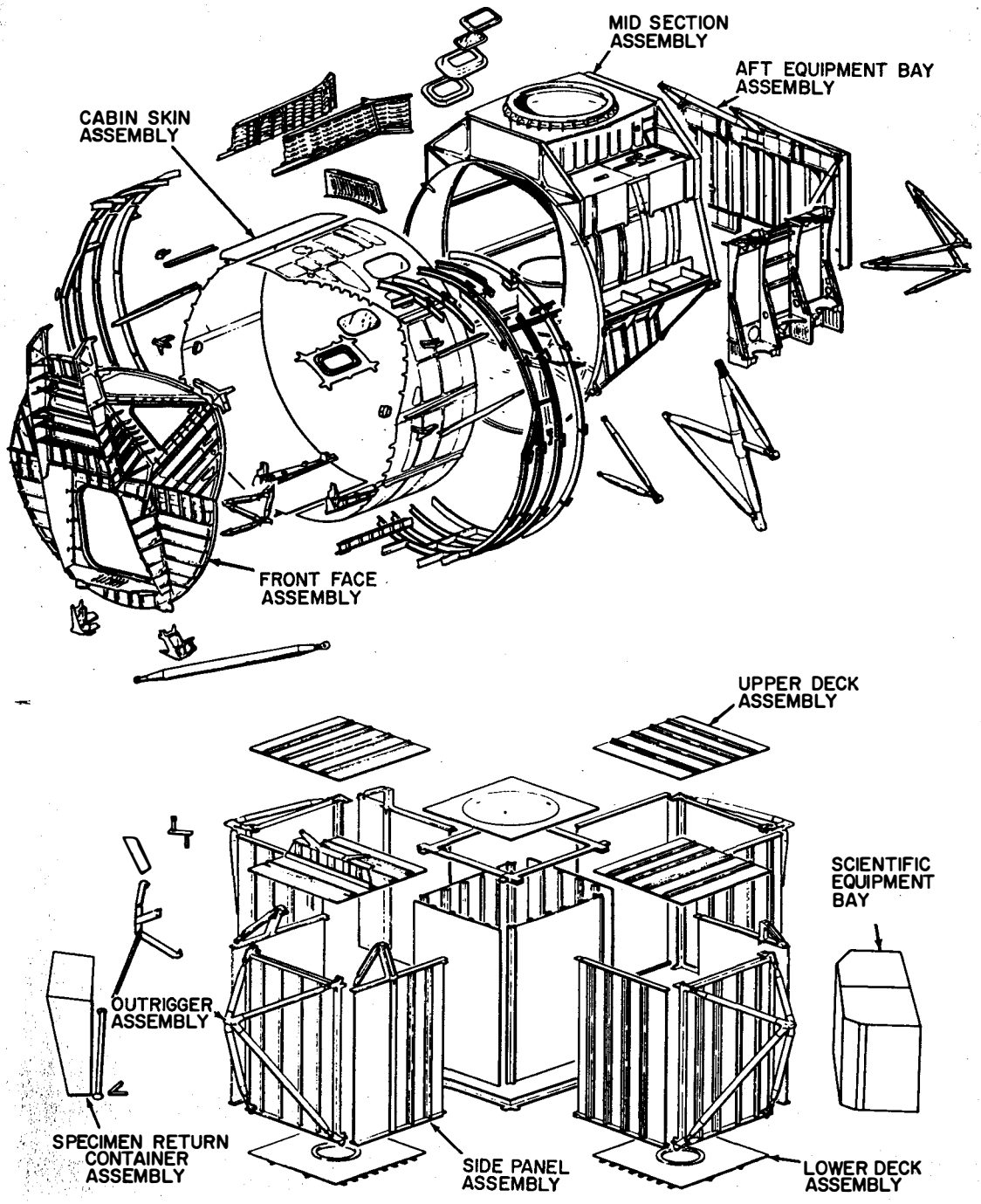
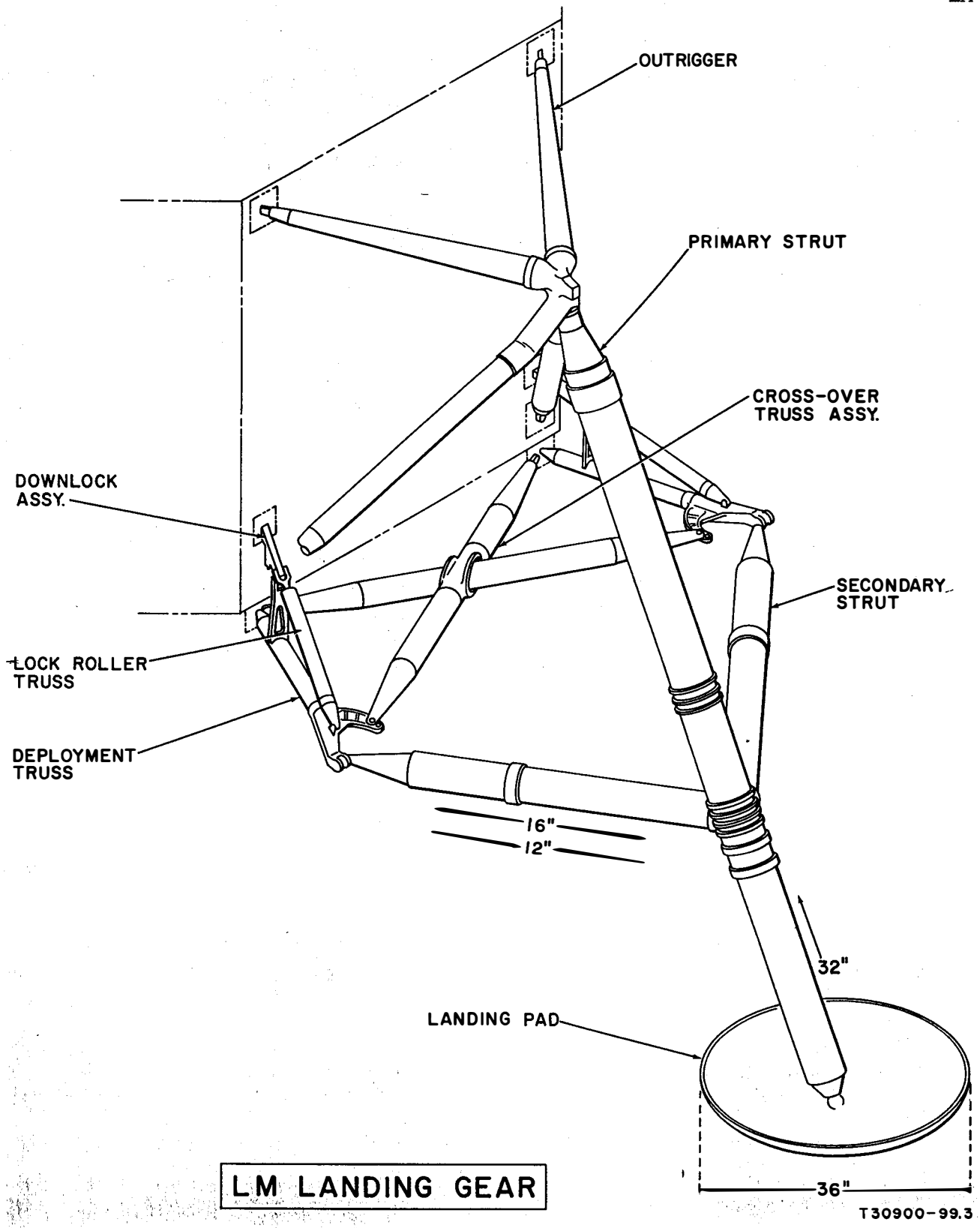



ASCENT AND DESCENT STAGE STRUCTURE

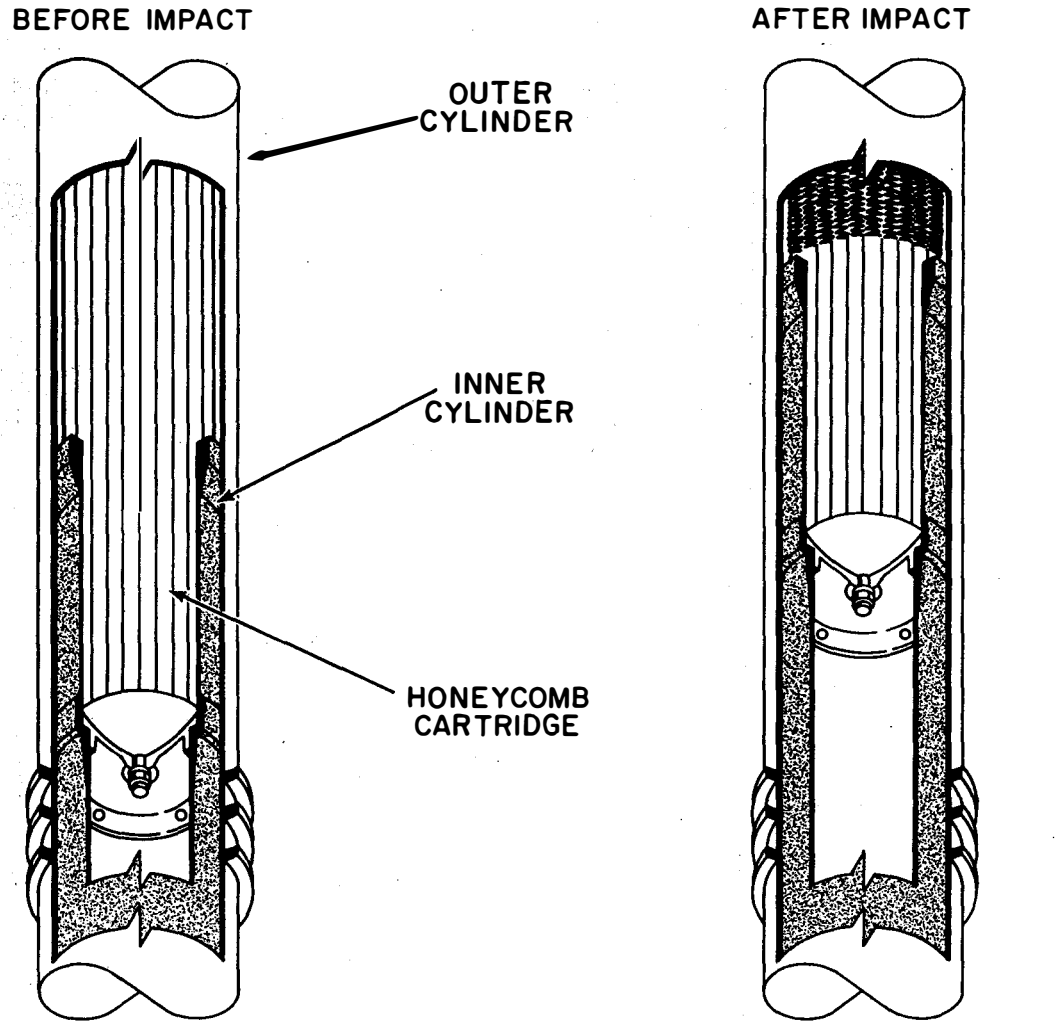
T30915-83.1

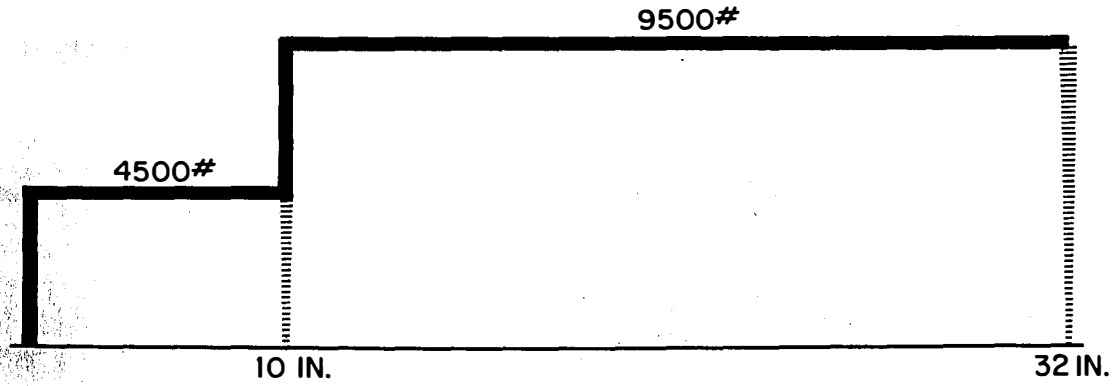


FOR TRAINING PURPOSES ONLY



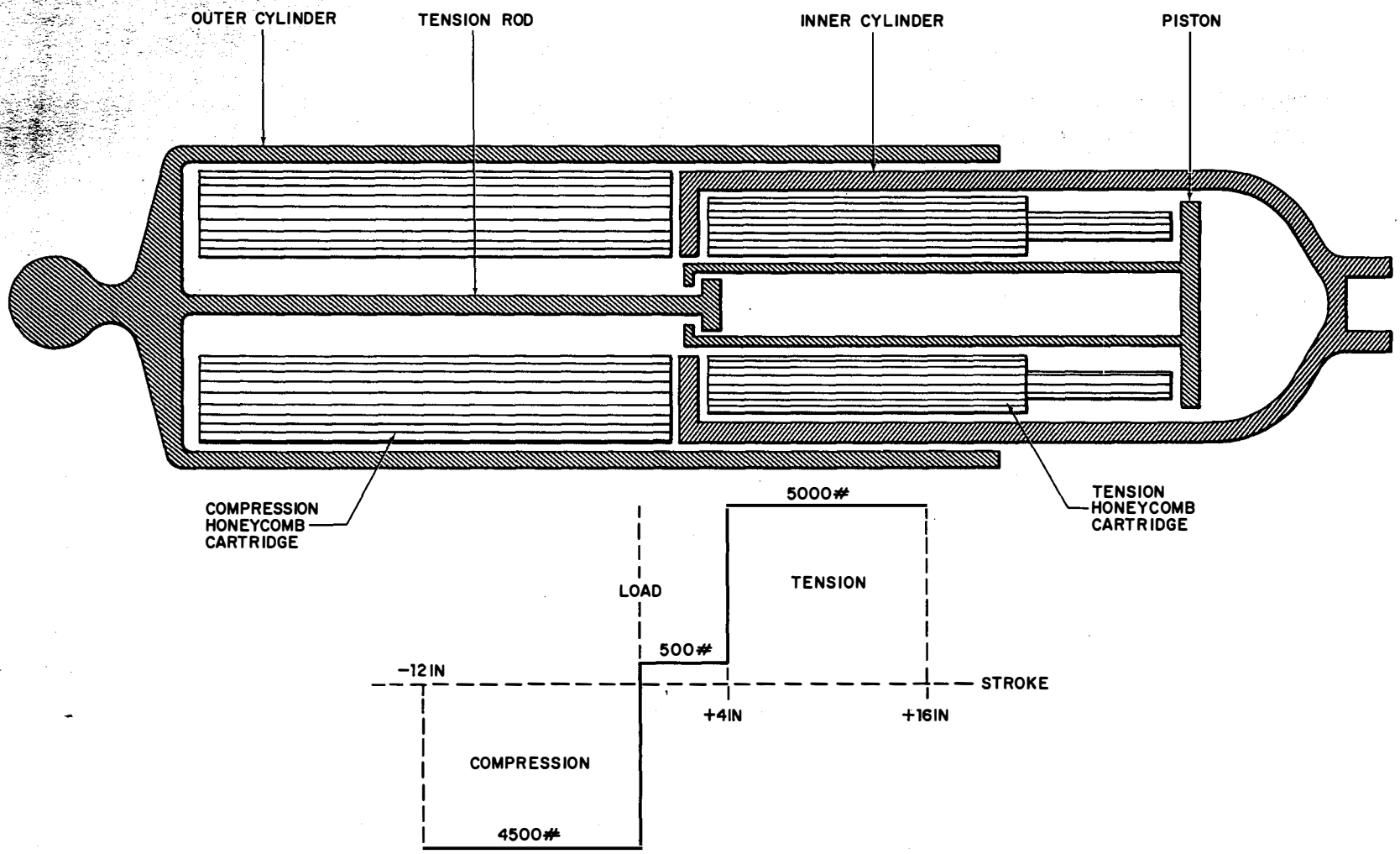

## LM LANDING GEAR




LANDING GEAR PRIMARY STRUT

T 30005-213




## LANDING GEAR SECONDARY STRUT

T30005-214

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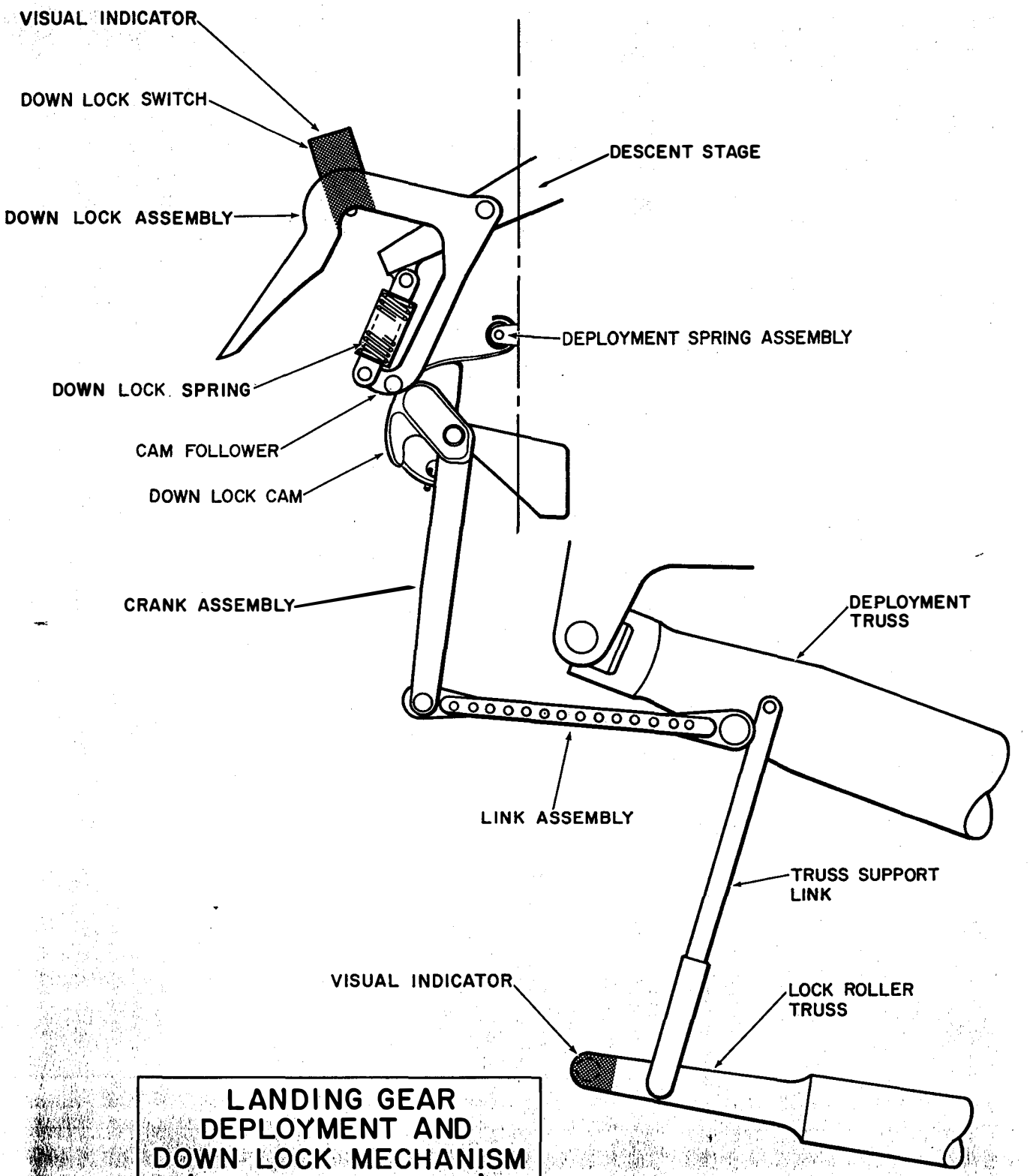
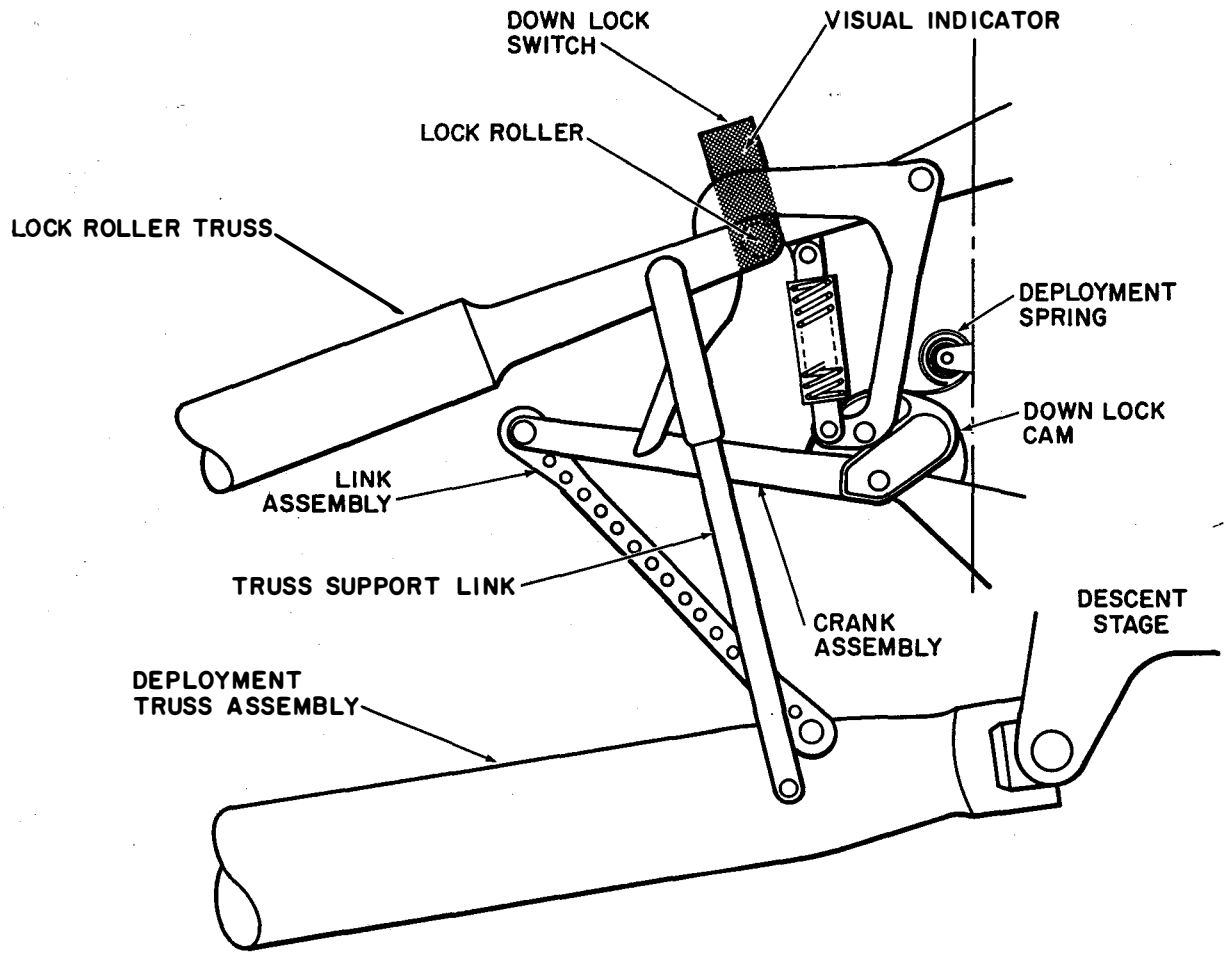
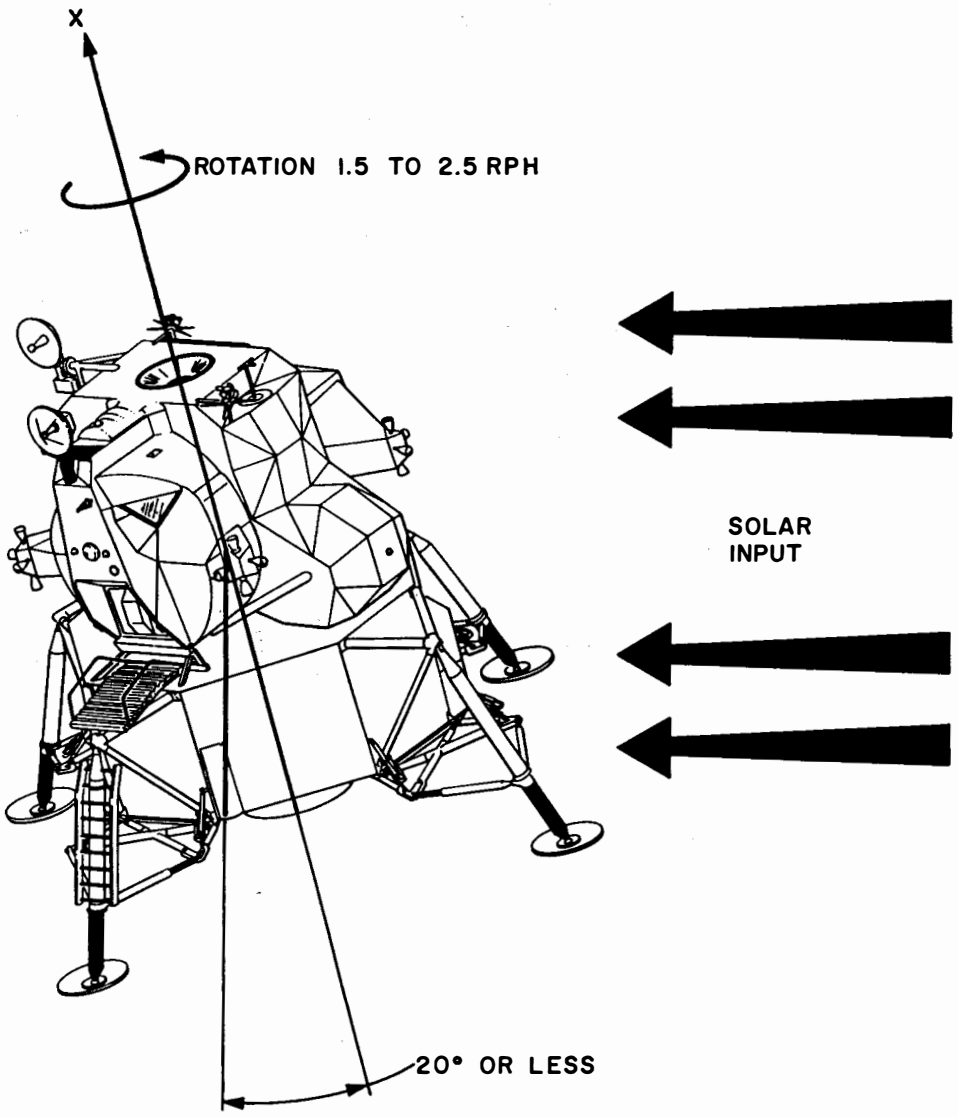



# LANDING GEAR DEPLOYMENT AND DOWN LOCK MECHANISM (STOWED POSITION)



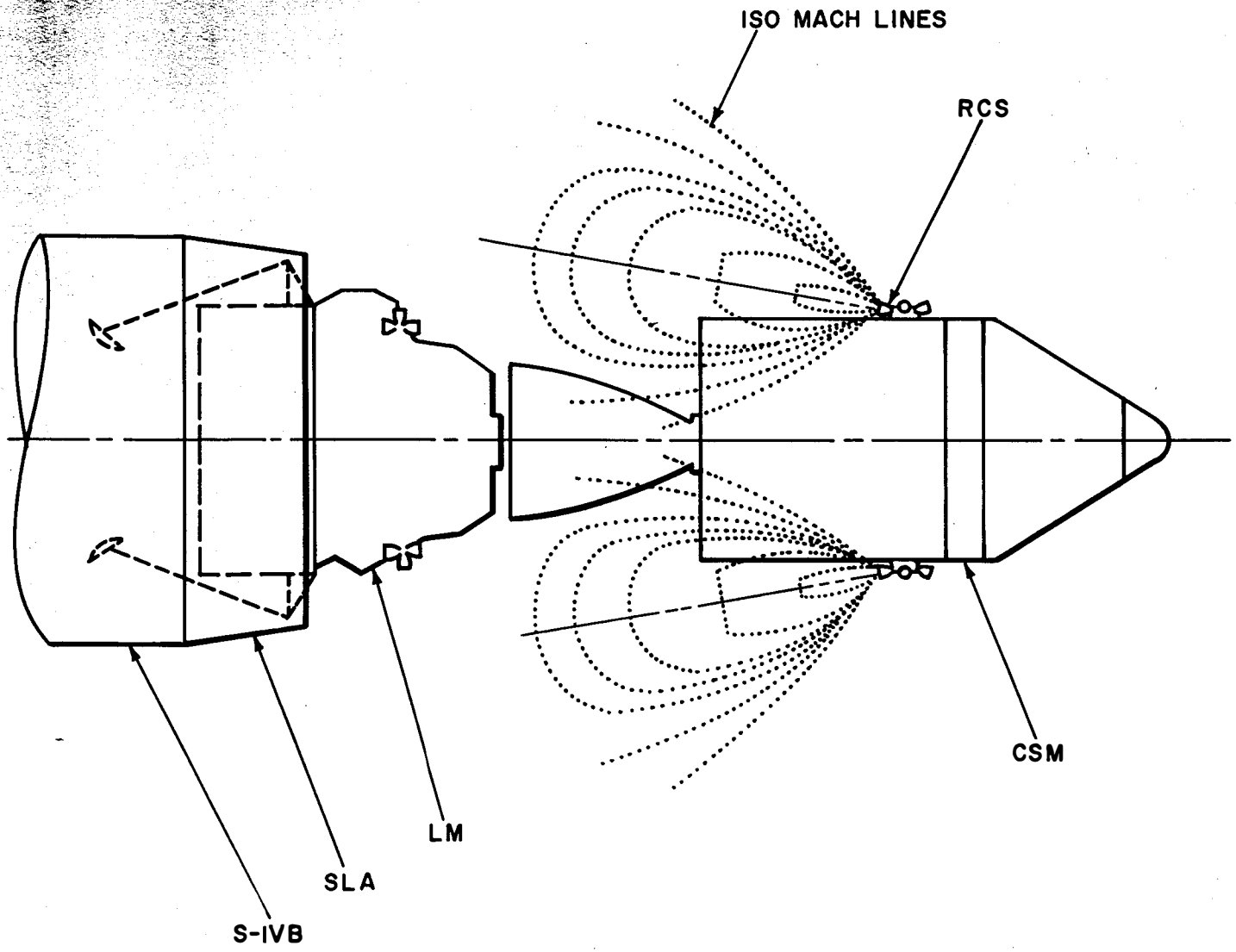

**LANDING GEAR DEPLOYMENT AND DOWN LOCK MECHANISM (DOWN AND LOCKED POSITION)**

T30915-19.2




## TRANSLUNAR FLIGHT

T30005-201.1




CSM SEPARATION FROM LM & S-IVB

T30915-96.1

IM-5

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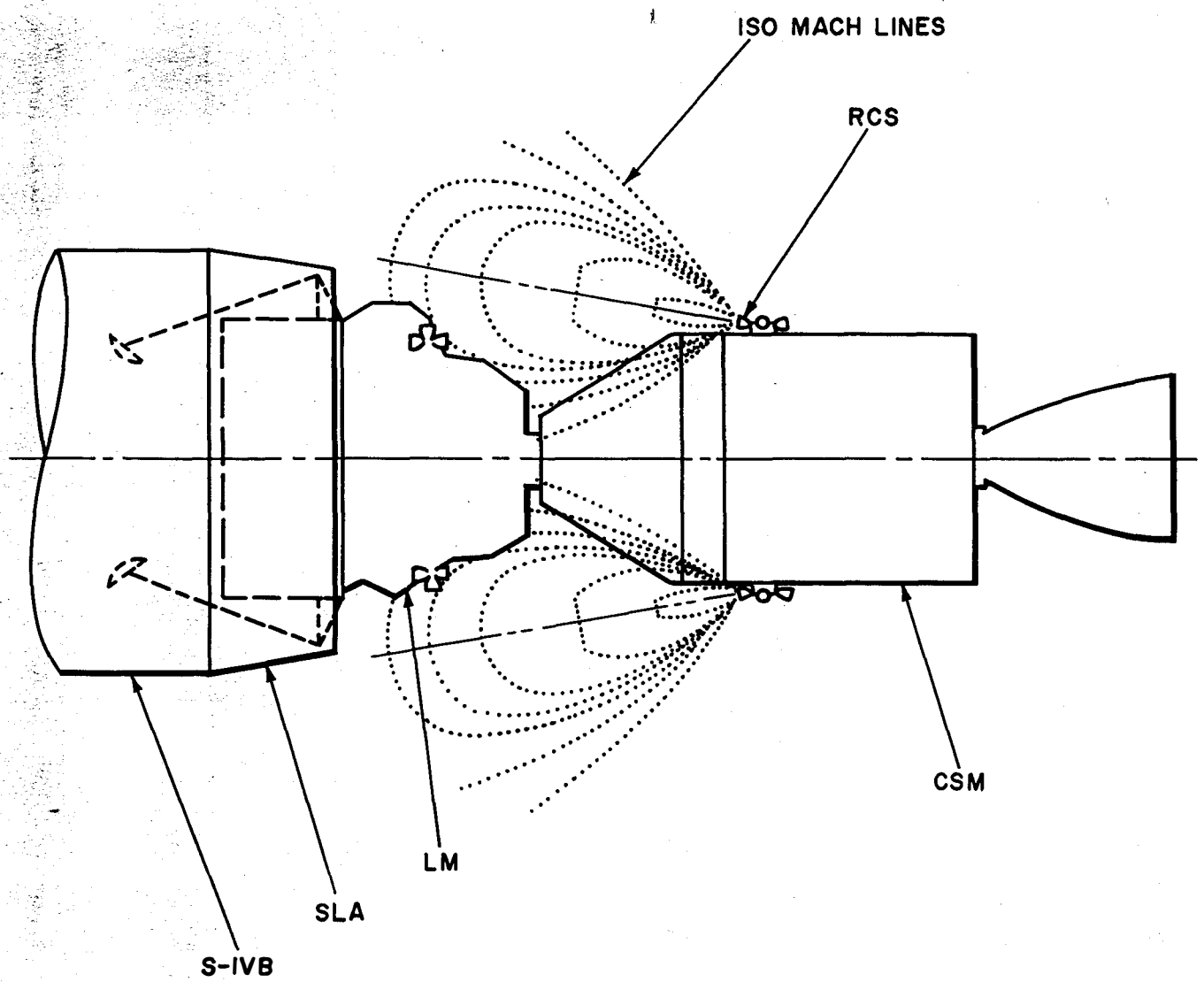


POST TRANSPOSITION DOCKING

T30915-98.1



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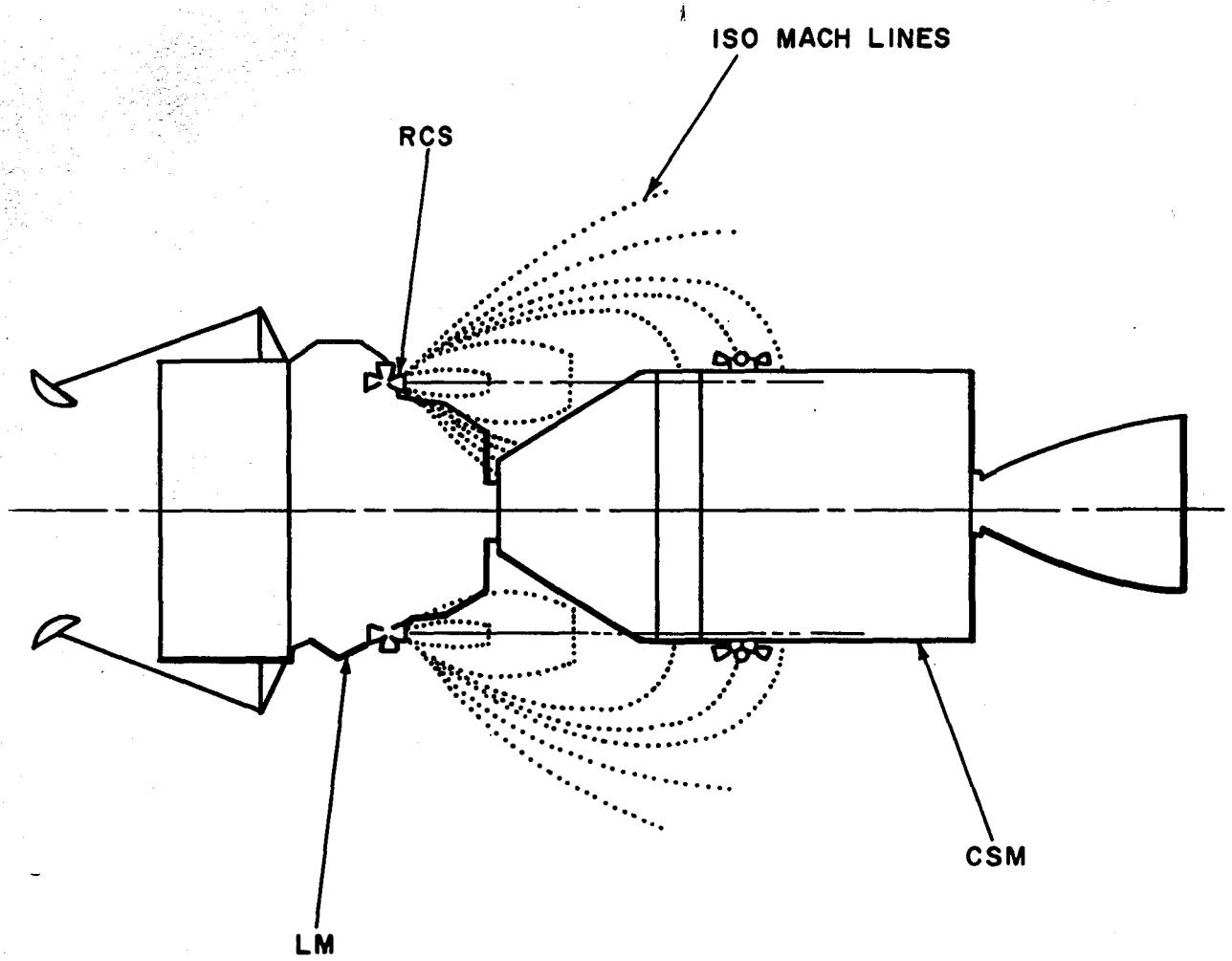


LM/CSM UNDOCKING

T30915-97.1

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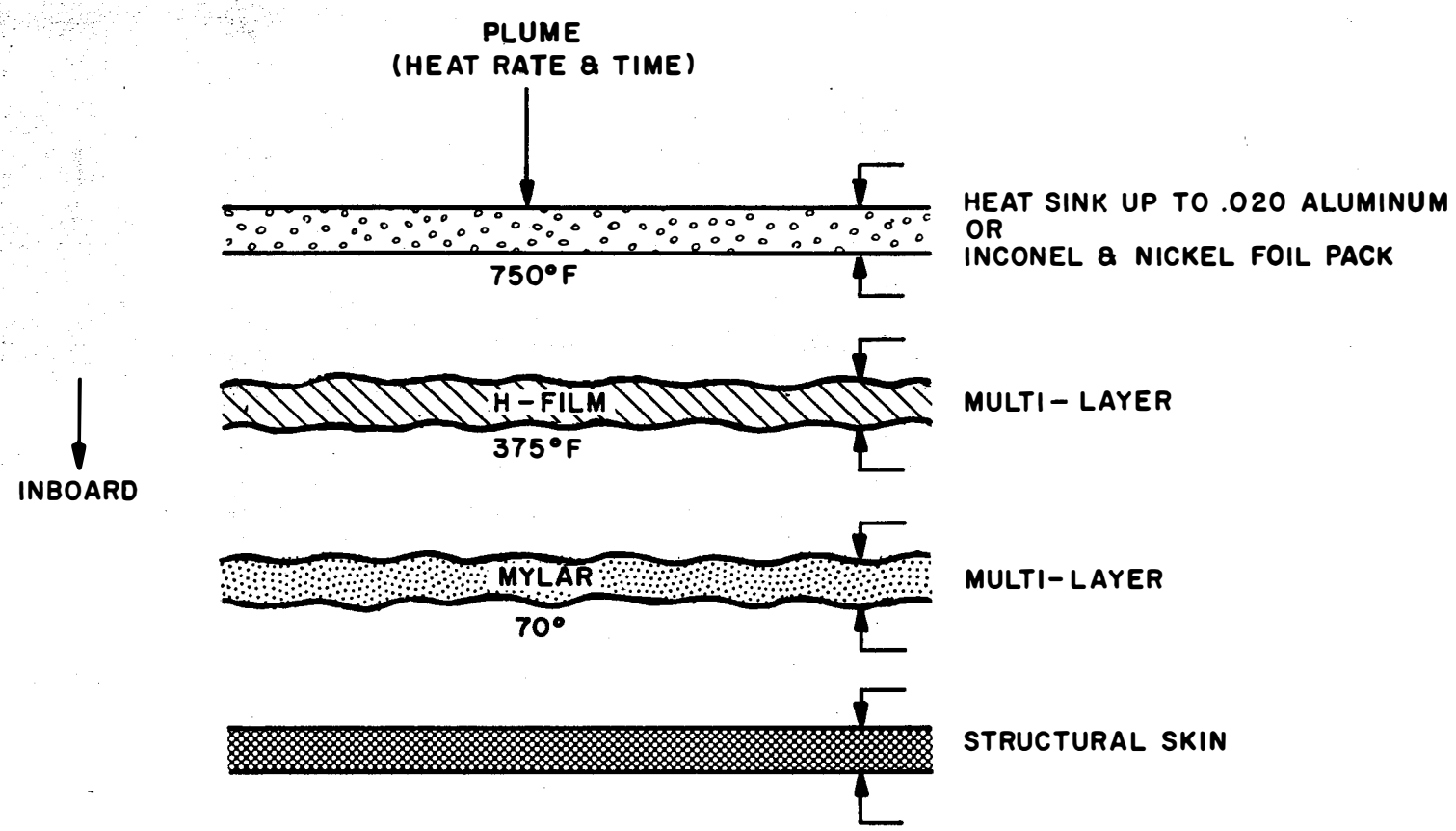


## THERMAL CONTROL

T30005-198.2

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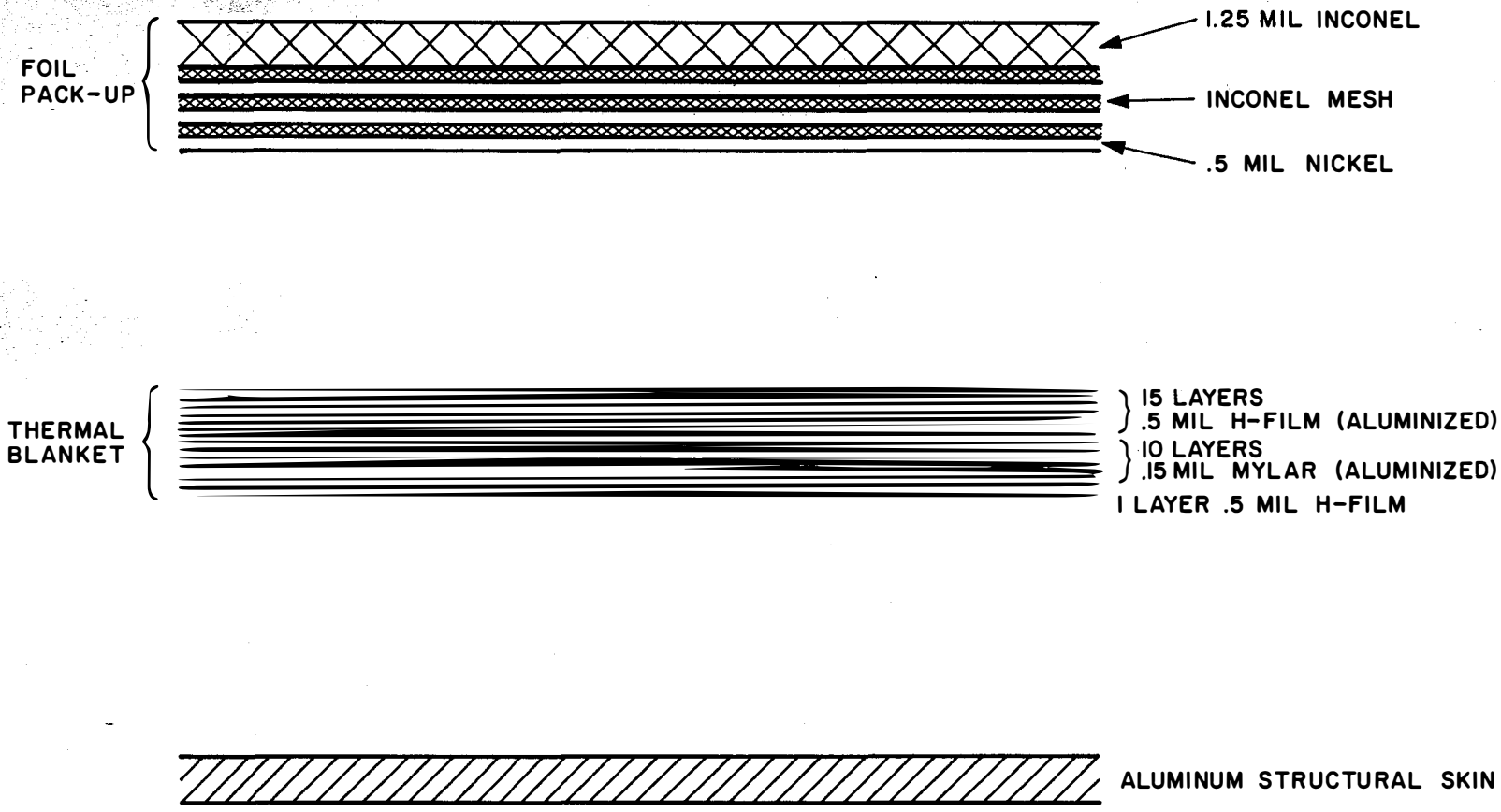
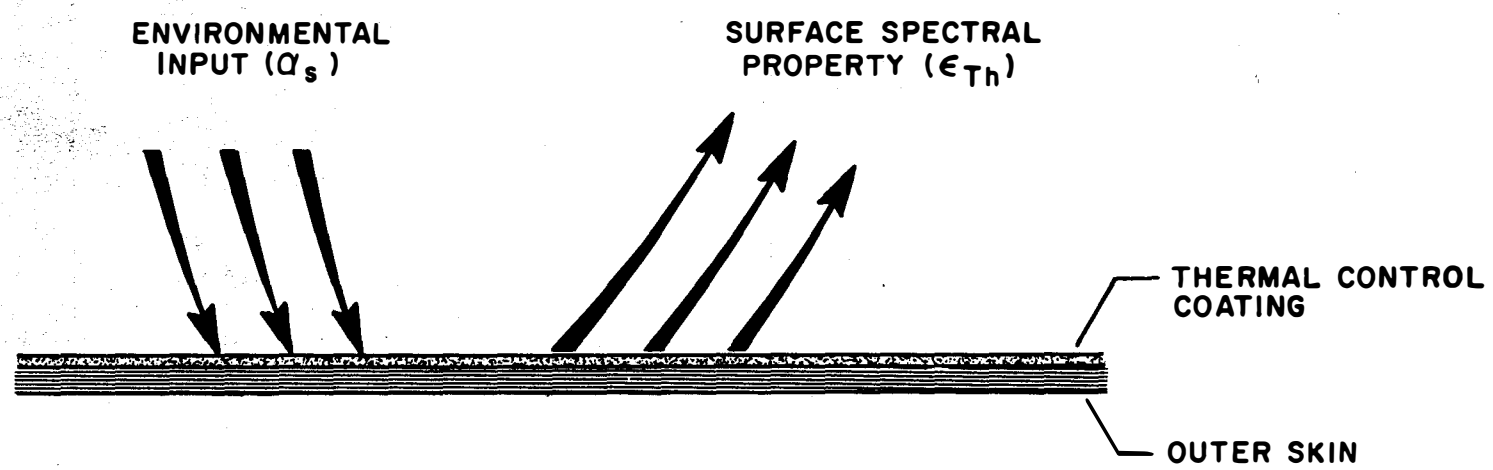



TYPICAL THERMAL SHIELD AND BLANKET

T30005-211.1




$\alpha_s$ = % OF THE TOTAL SOLAR ENERGY ABSORBED BY A SURFACE. (0 – 1.0)

$\epsilon_{th}$ = ABILITY OF A SURFACE TO REJECT HEAT TO SPACE. (0 – 1.0)

## THERMAL CONTROL COATING

T30005-204.1



Date: May 1969
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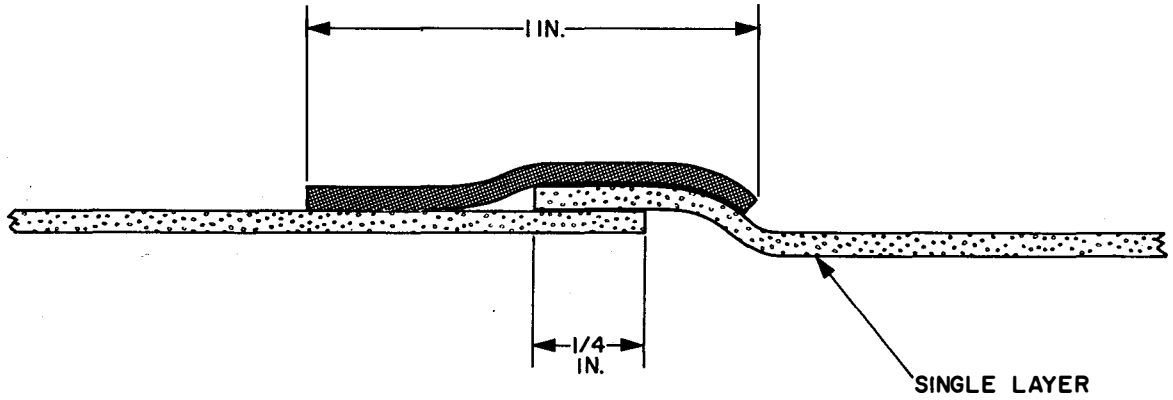
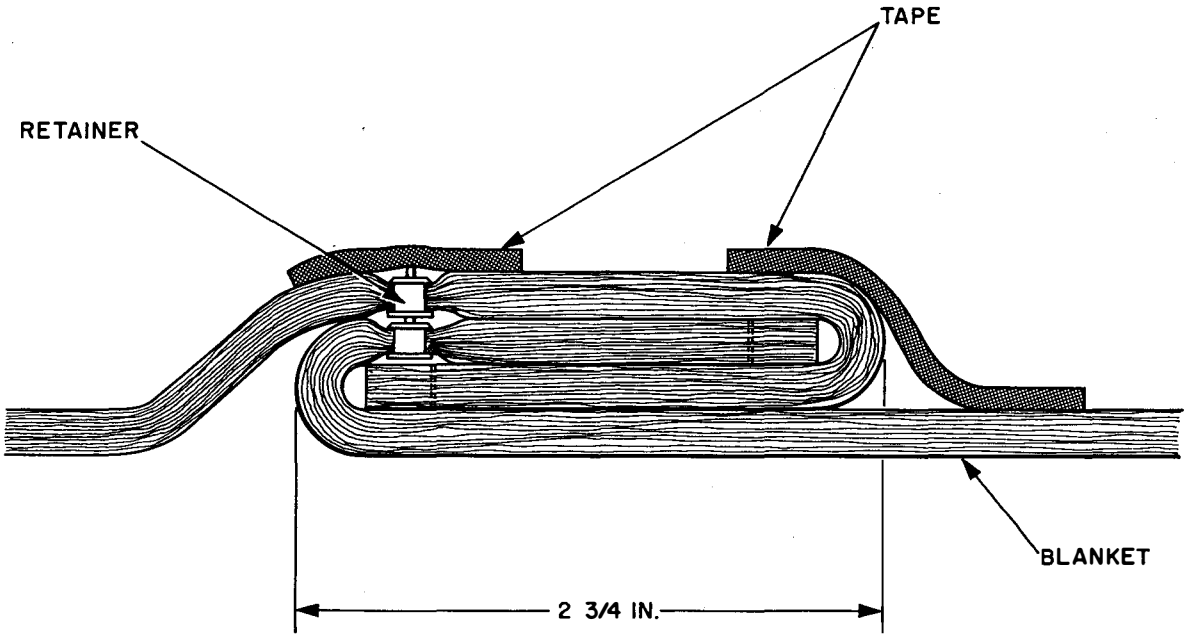
Page 1-13




TAPED SEAM




FOLDED SEAM

## BLANKET JOINING

T30005-205.1

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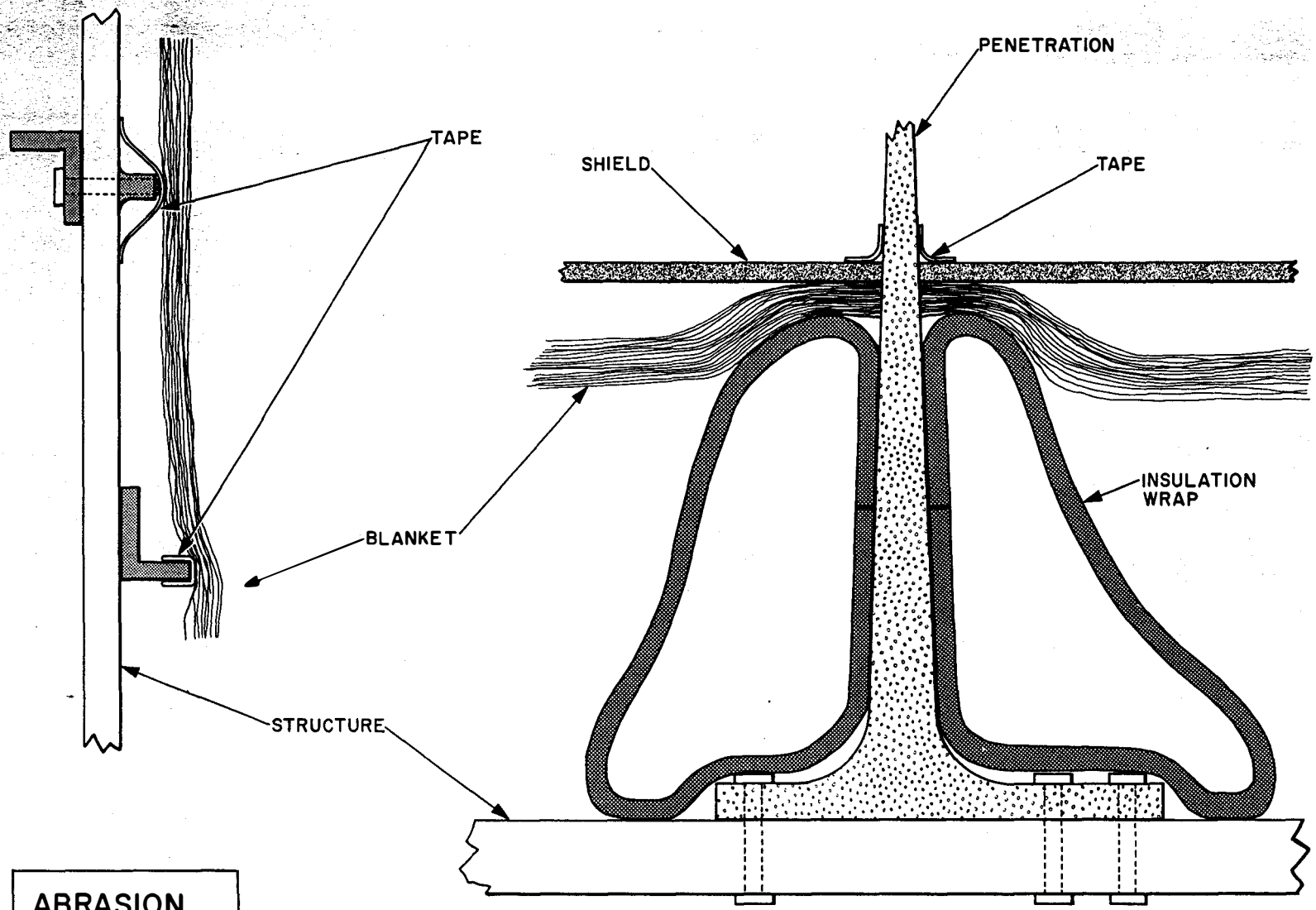

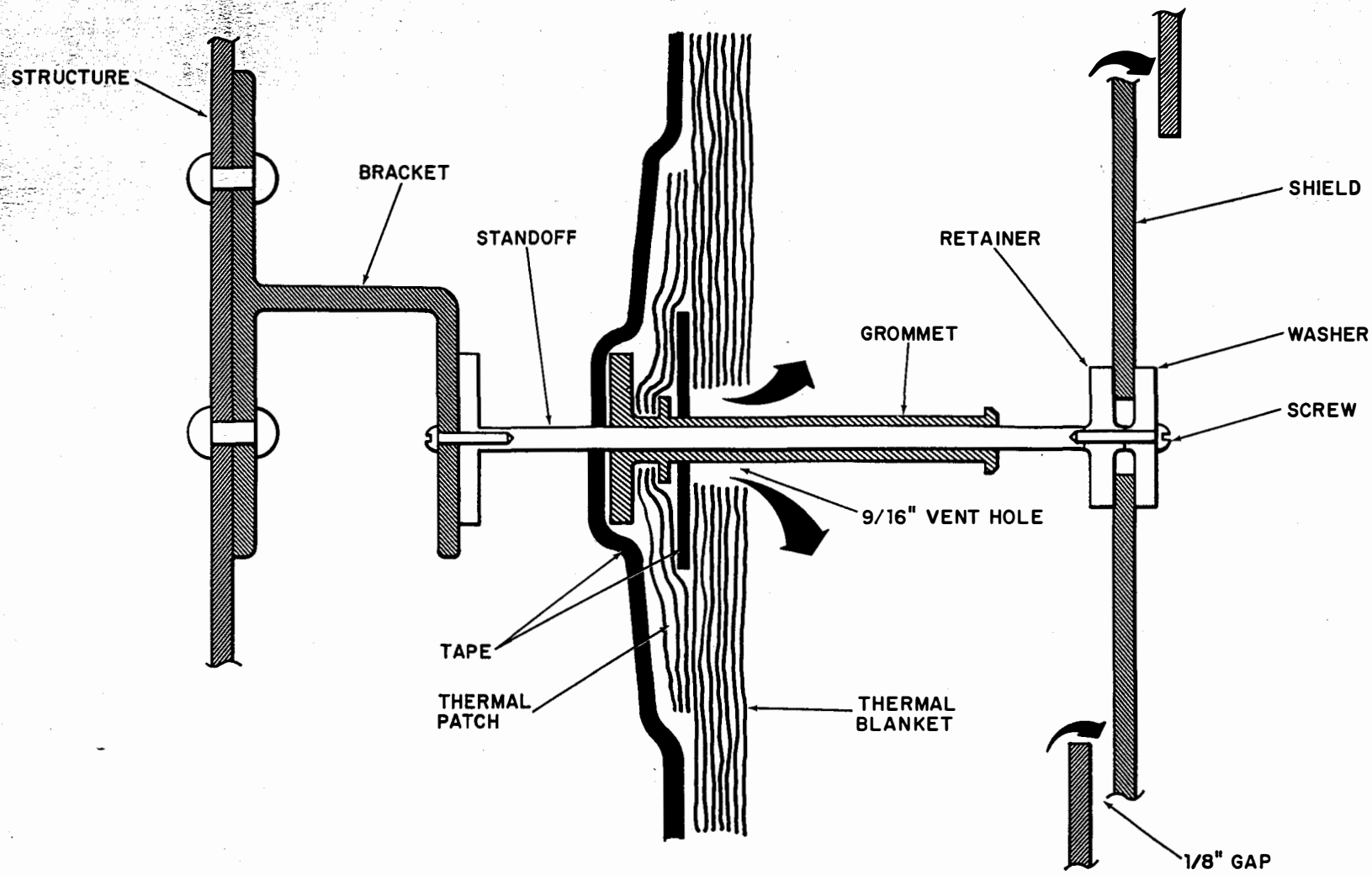


## STANDOFF INSTALLATION AND BLANKET VENTING

T30005-212

Date: May 1969
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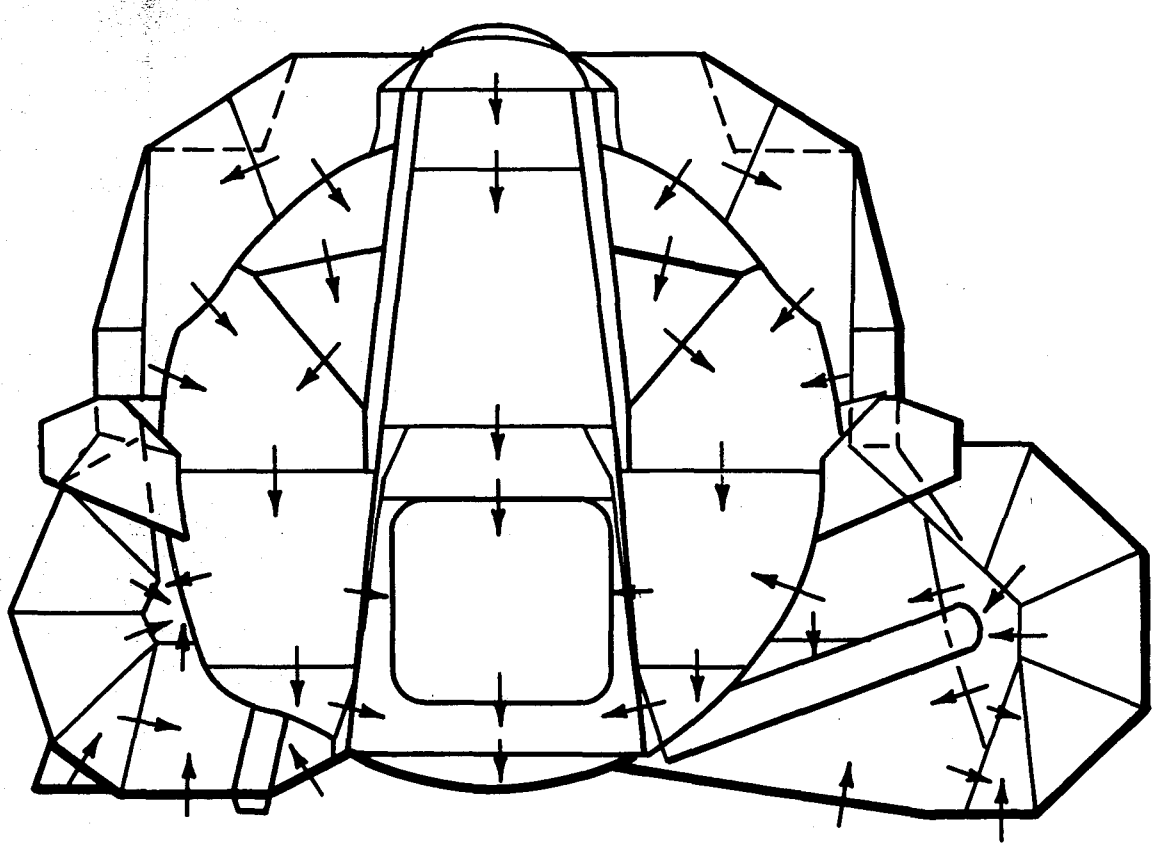
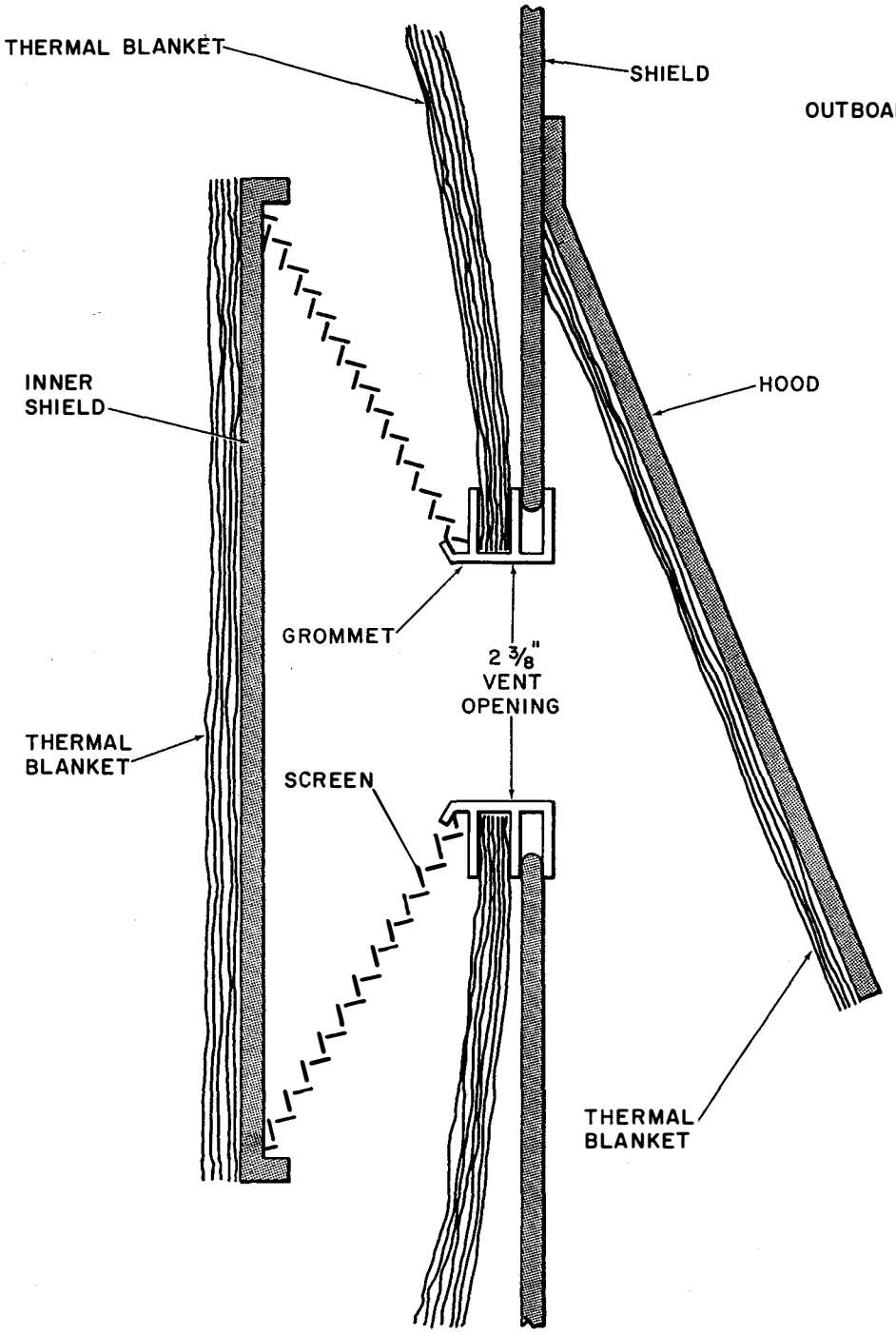


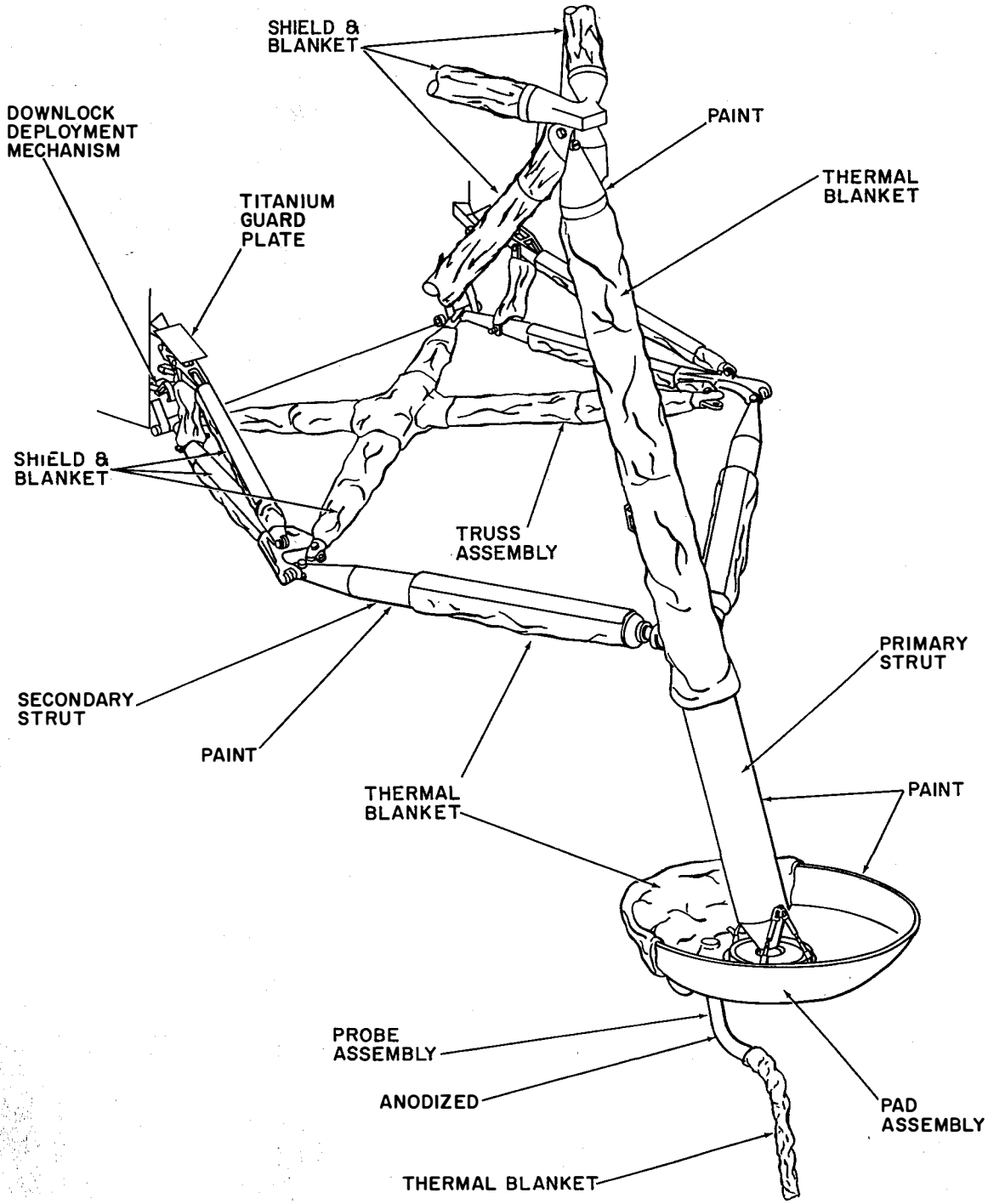
**PANEL SHINGLING
FRONT VIEW**

T30005-203.1




BAY VENT

T30005-215.1

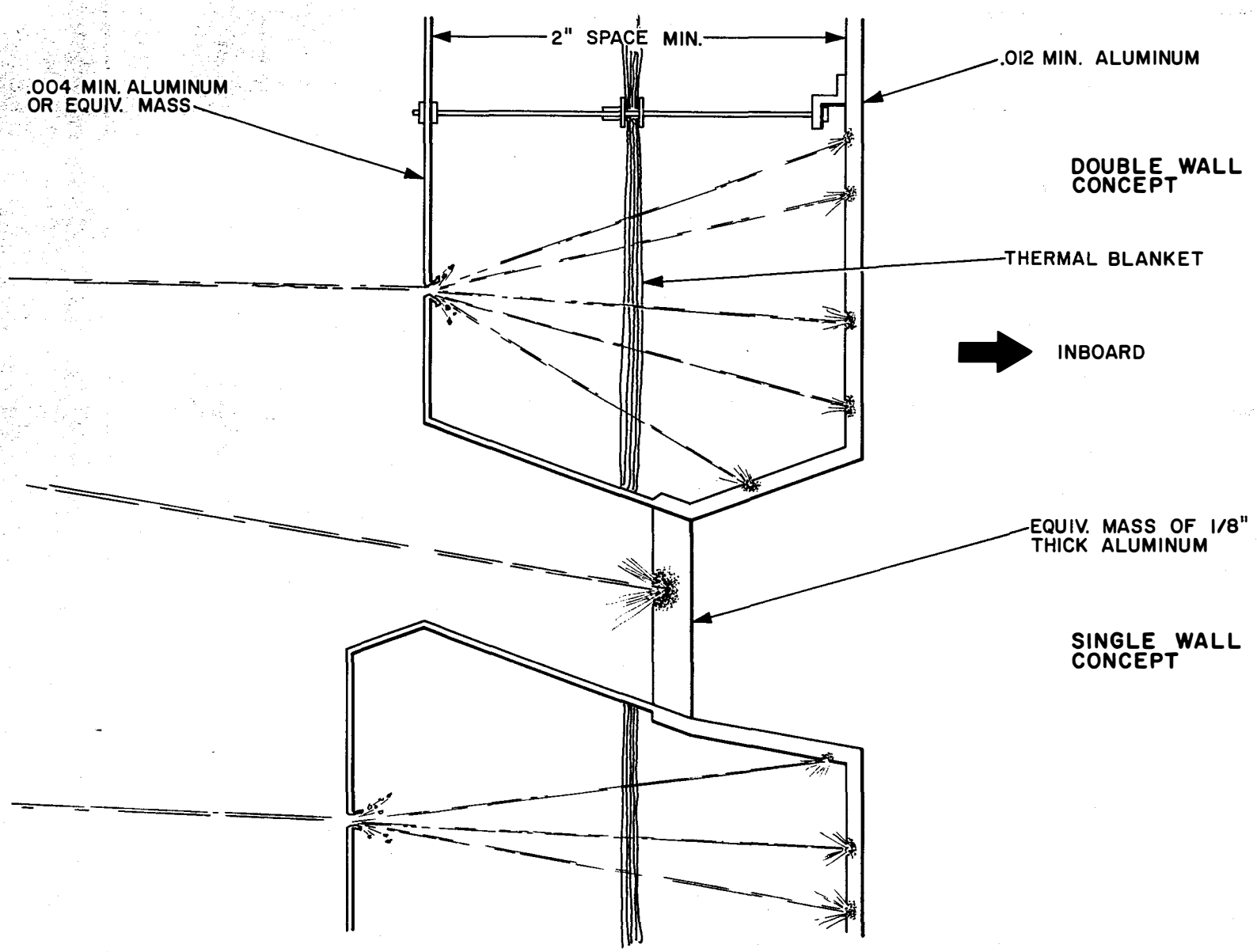



## LANDING GEAR THERMAL PROTECTION

T30915-101.2


FOR TRAINING PURPOSES ONLY

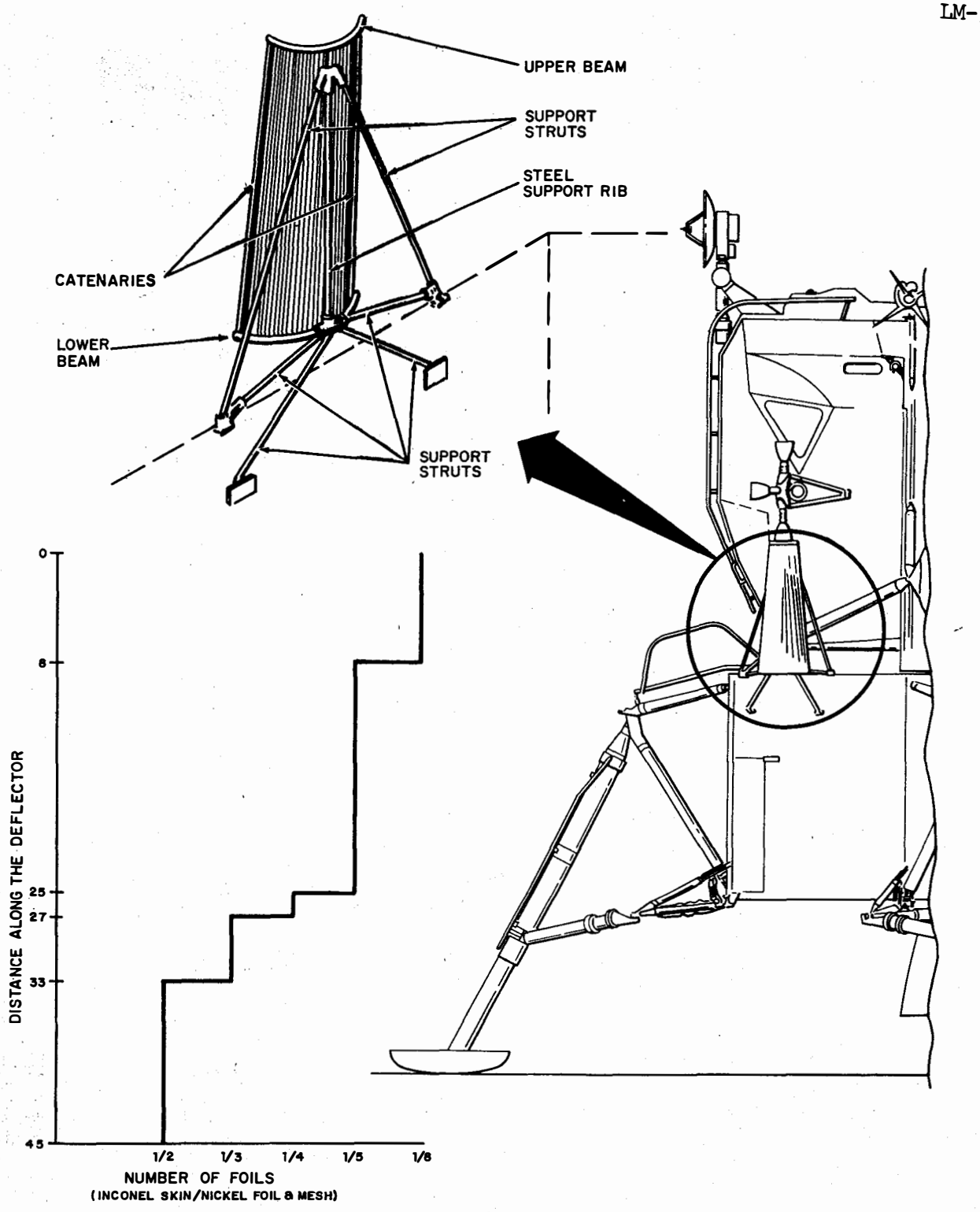
Date: May 1969
FOR TRAINING PURPOSES ONLY




## MICRO-METEOROID PROTECTION

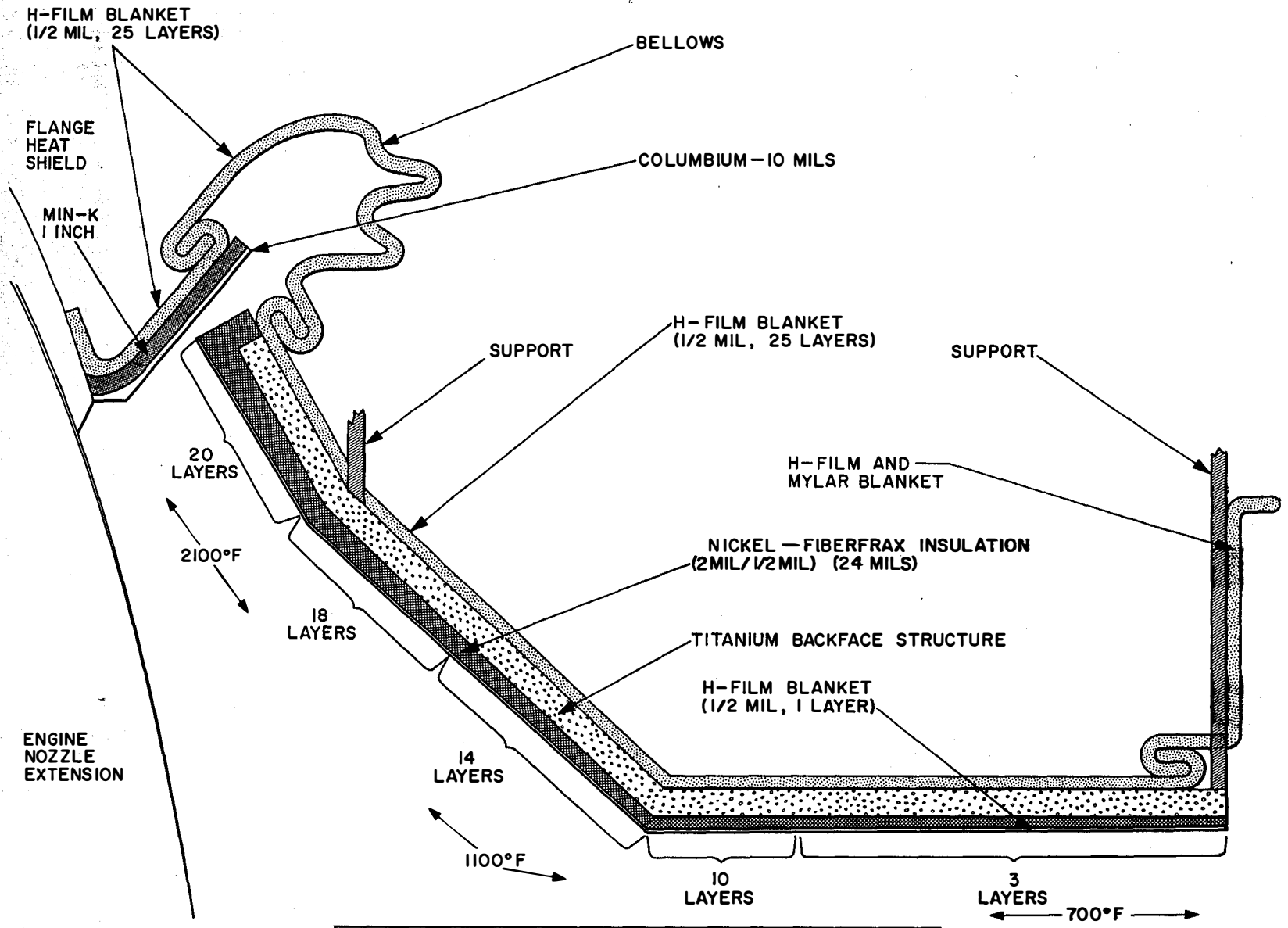
T30005-202.2

LM-5




RCS PLUME DEFLECTOR

T30005-216




DESCENT STAGE BASE HEAT SHIELD

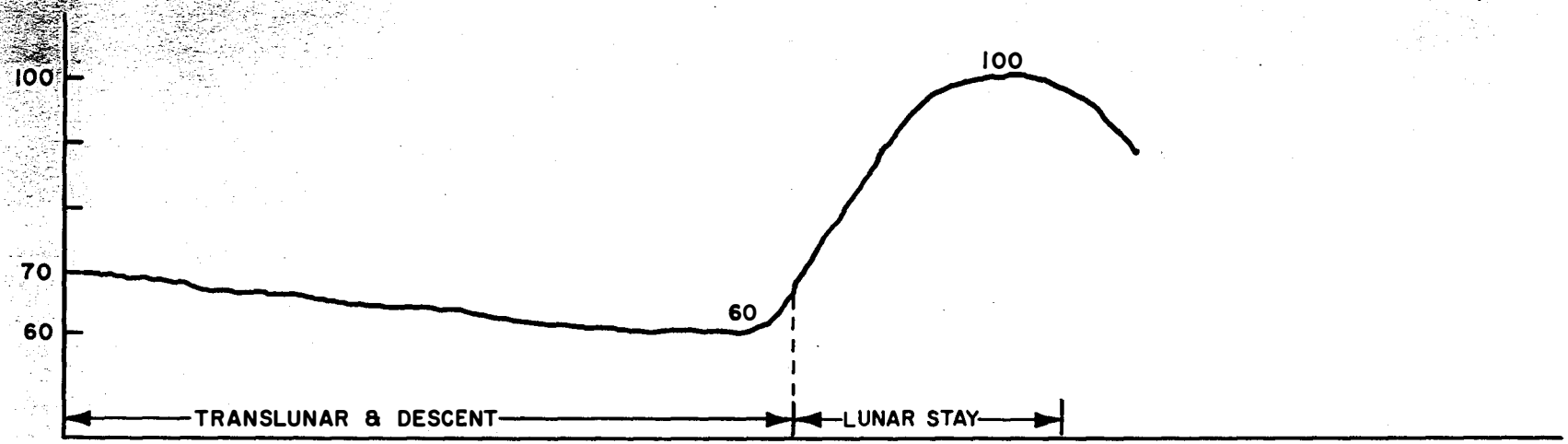
T30005-196.2

LM-5

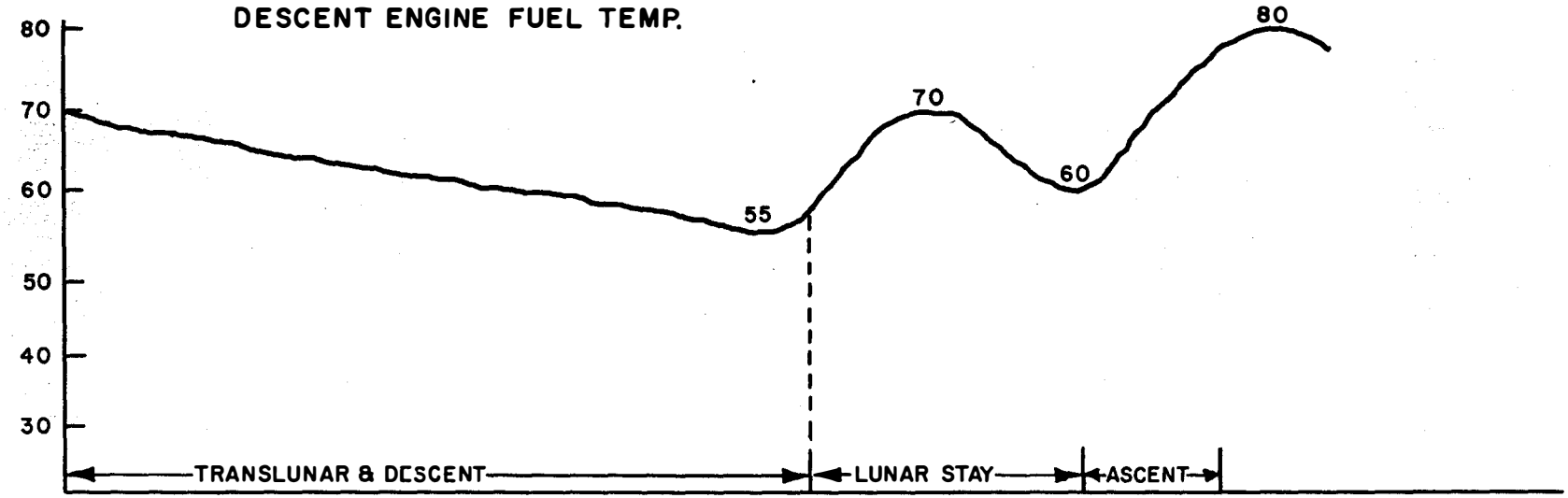


FOR TRAINING PURPOSES ONLY

Dated: May 1969
FOR TRAINING PURPOSES ONLY




DESCENT ENGINE FUEL TEMP.

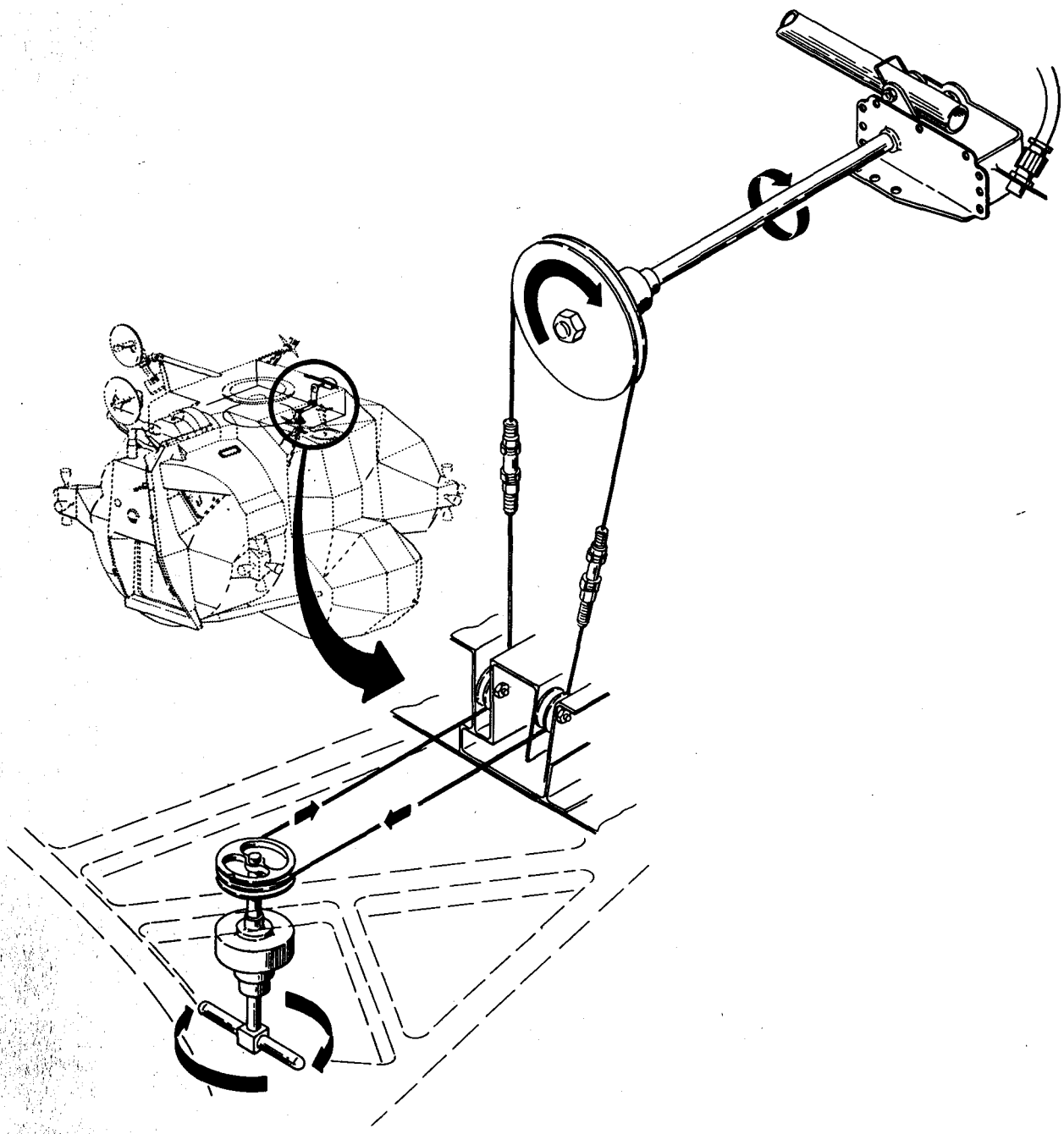



CABIN TEMP.

## TEMPERATURE GRAPHS

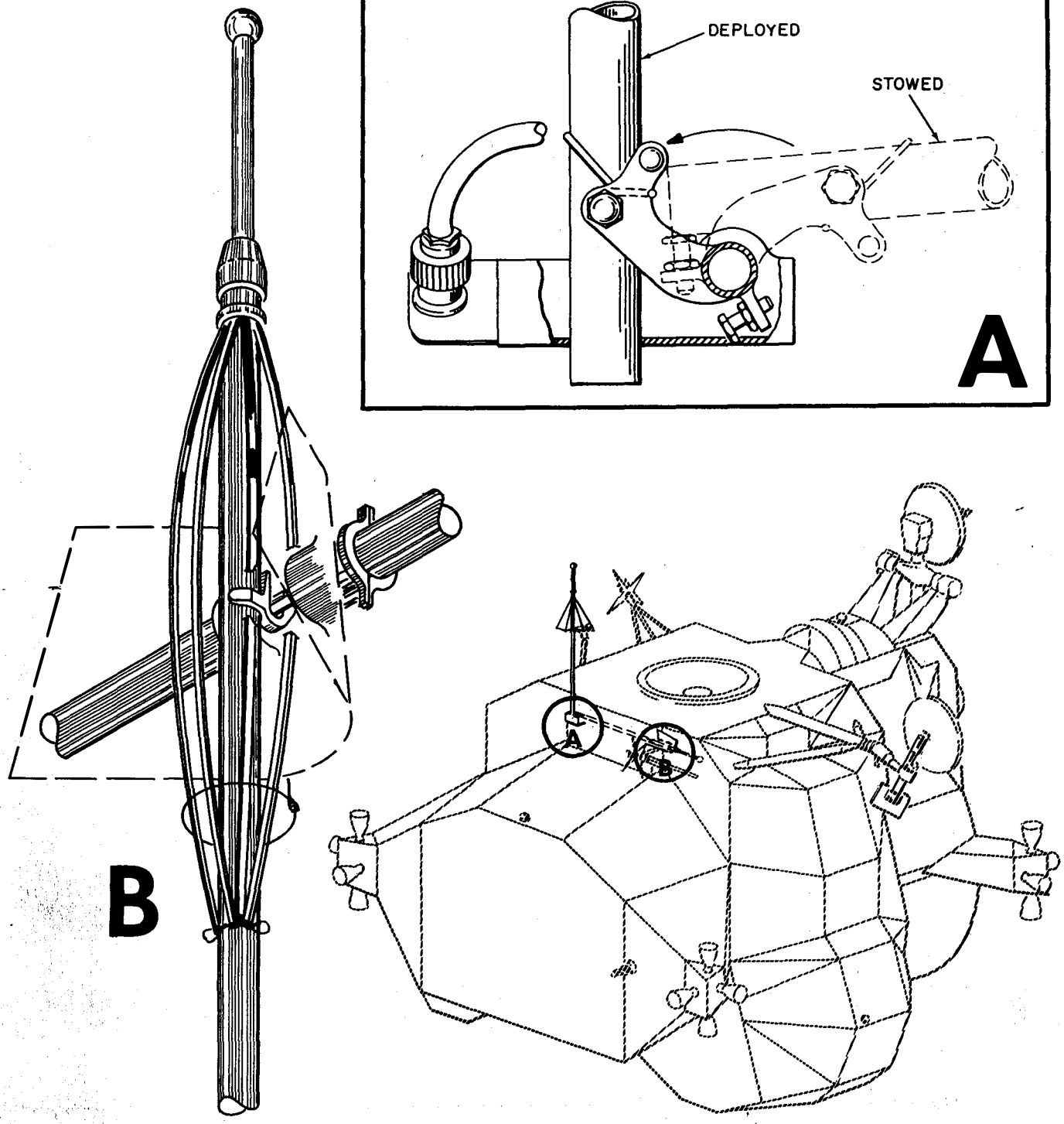
T30005-206.2

IM-5



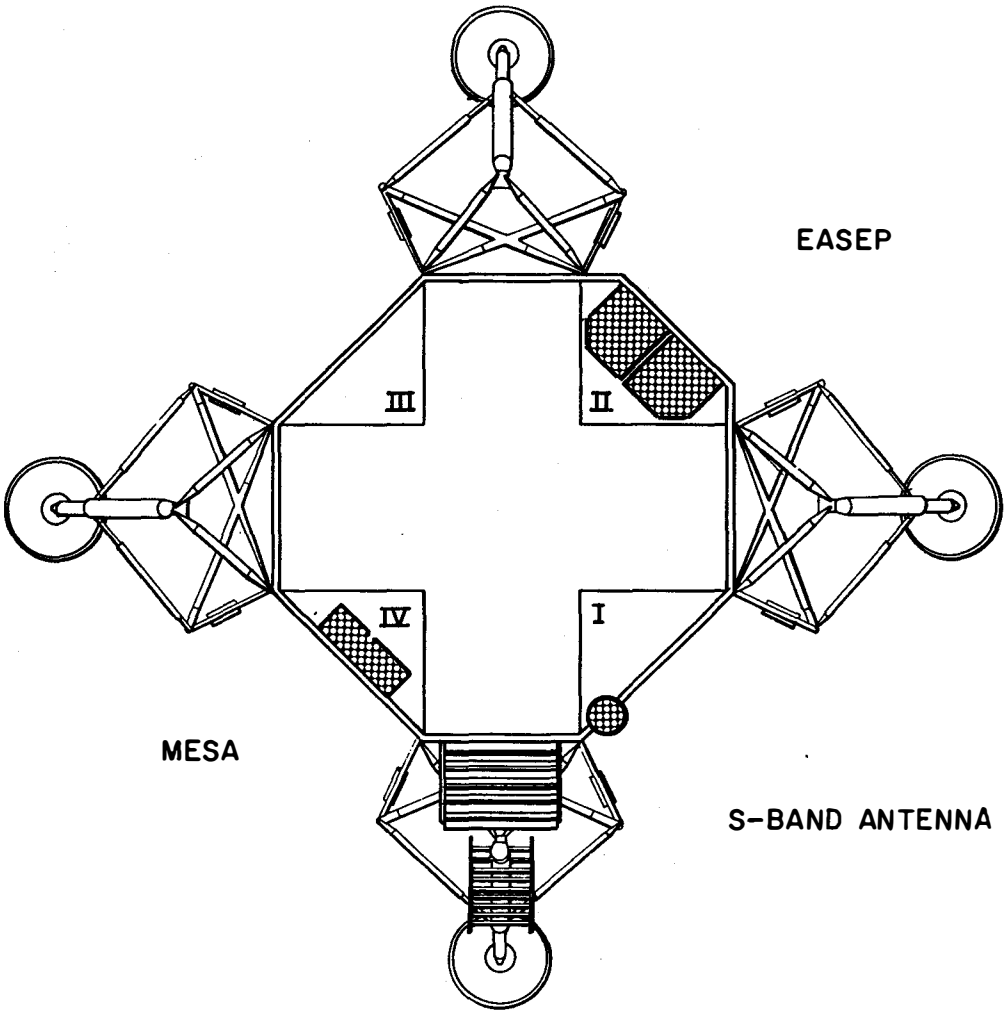
EVA ANTENNA ERECTION MECHANISM (1)

T30715-142



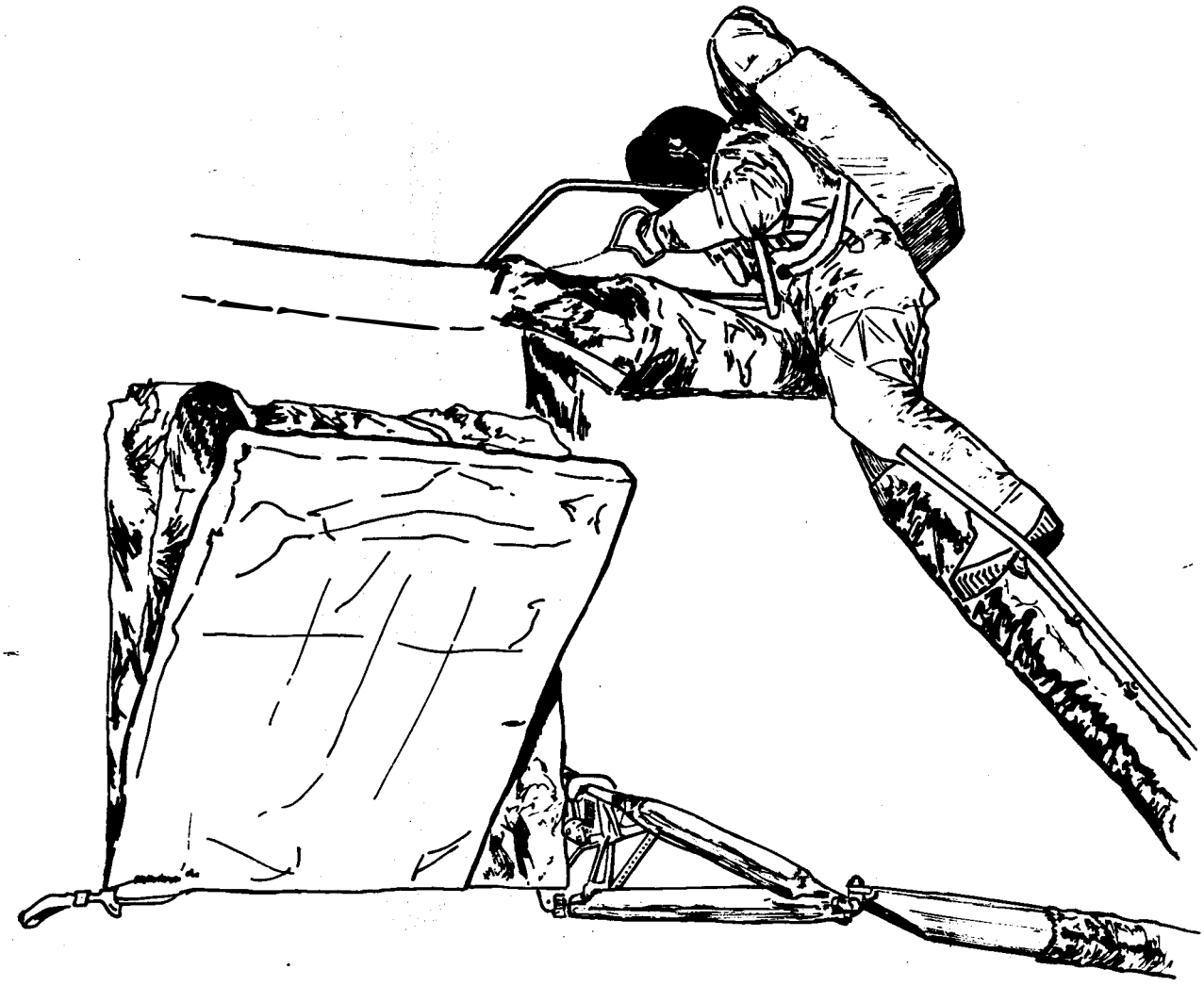

EVA ANTENNA ERECTION MECHANISM (2)

T30715-145



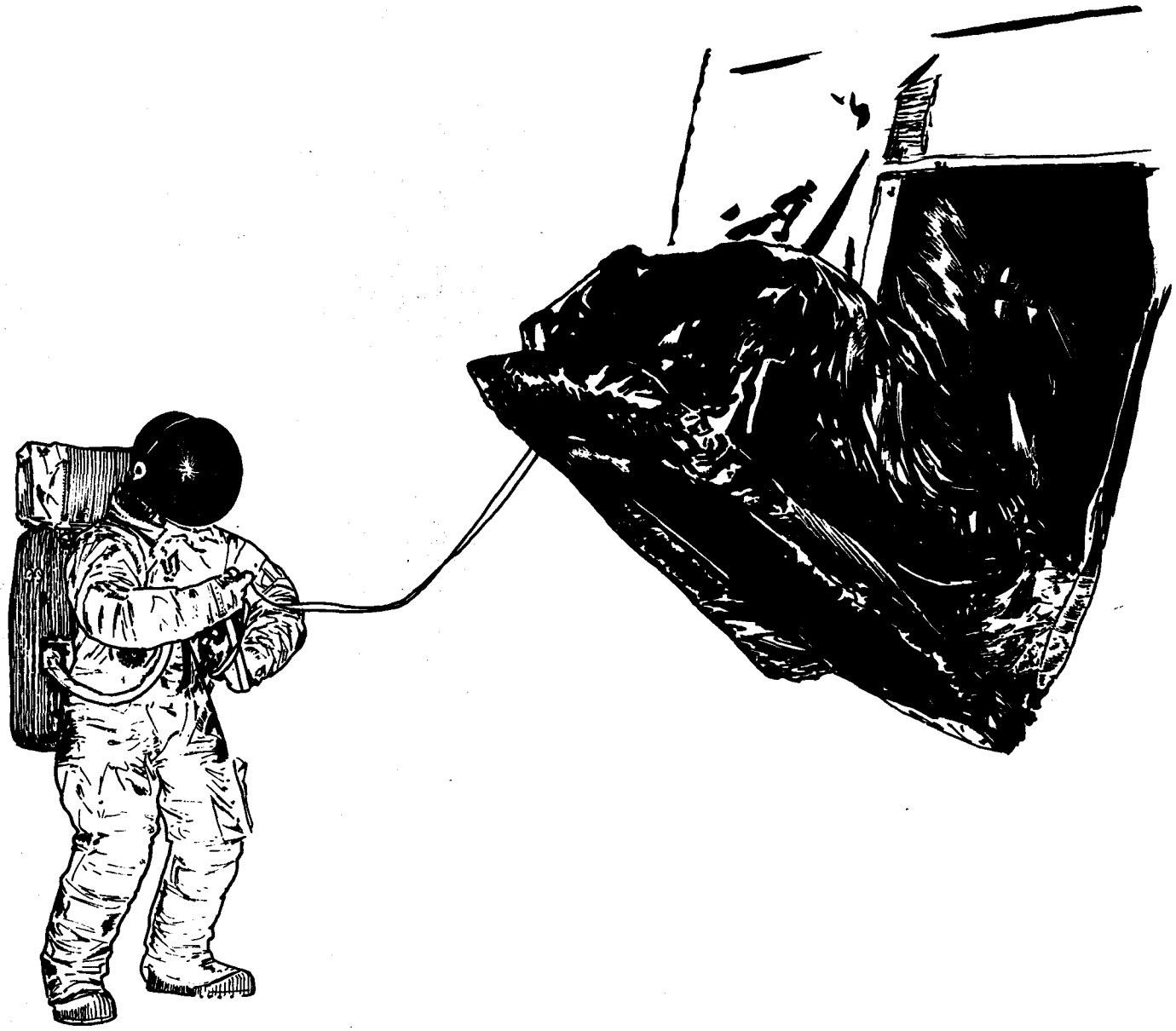

EQUIPMENT LOCATIONS - DESCENT STAGE

T30005-220



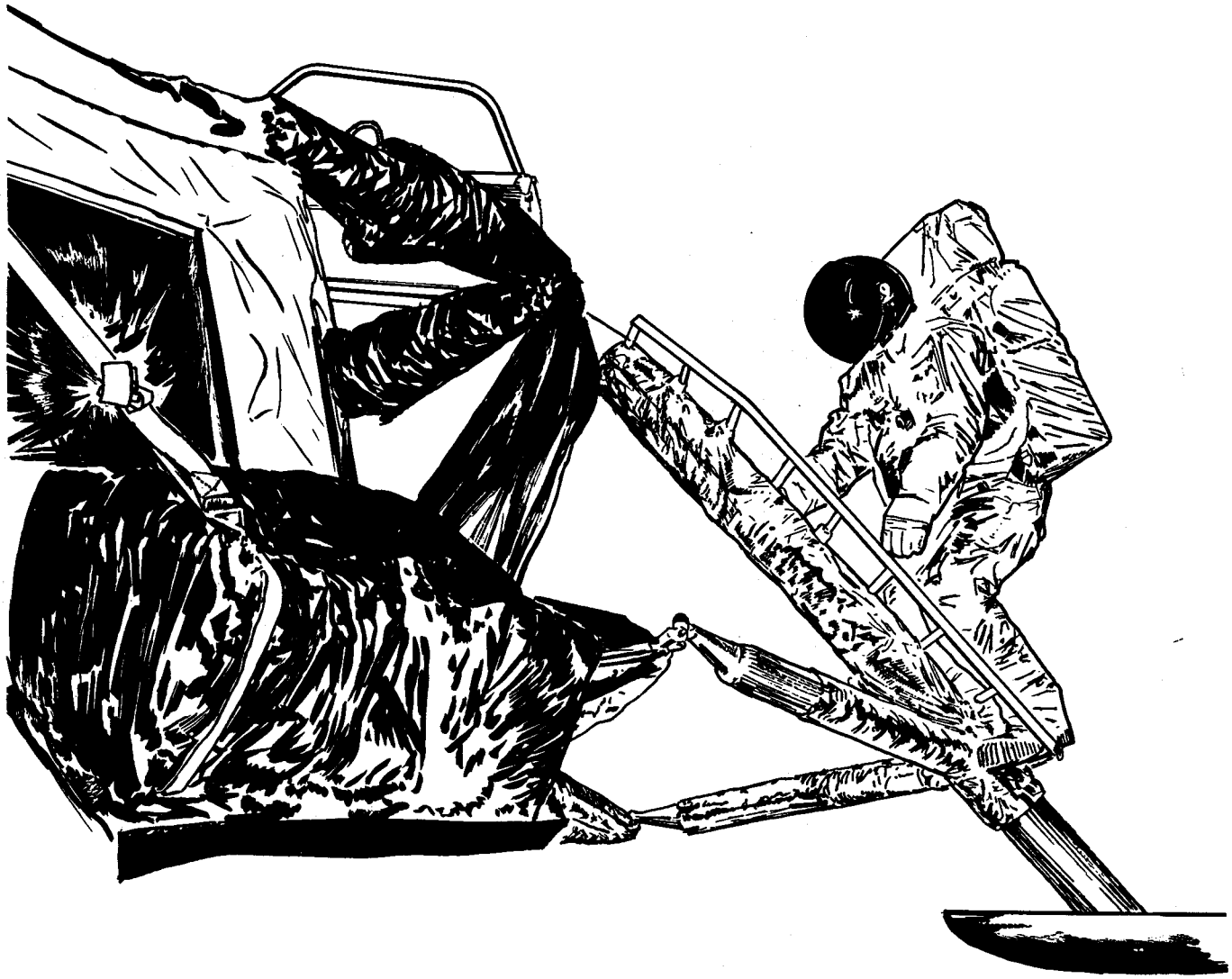
MESA DEPLOYMENT

T30005-222



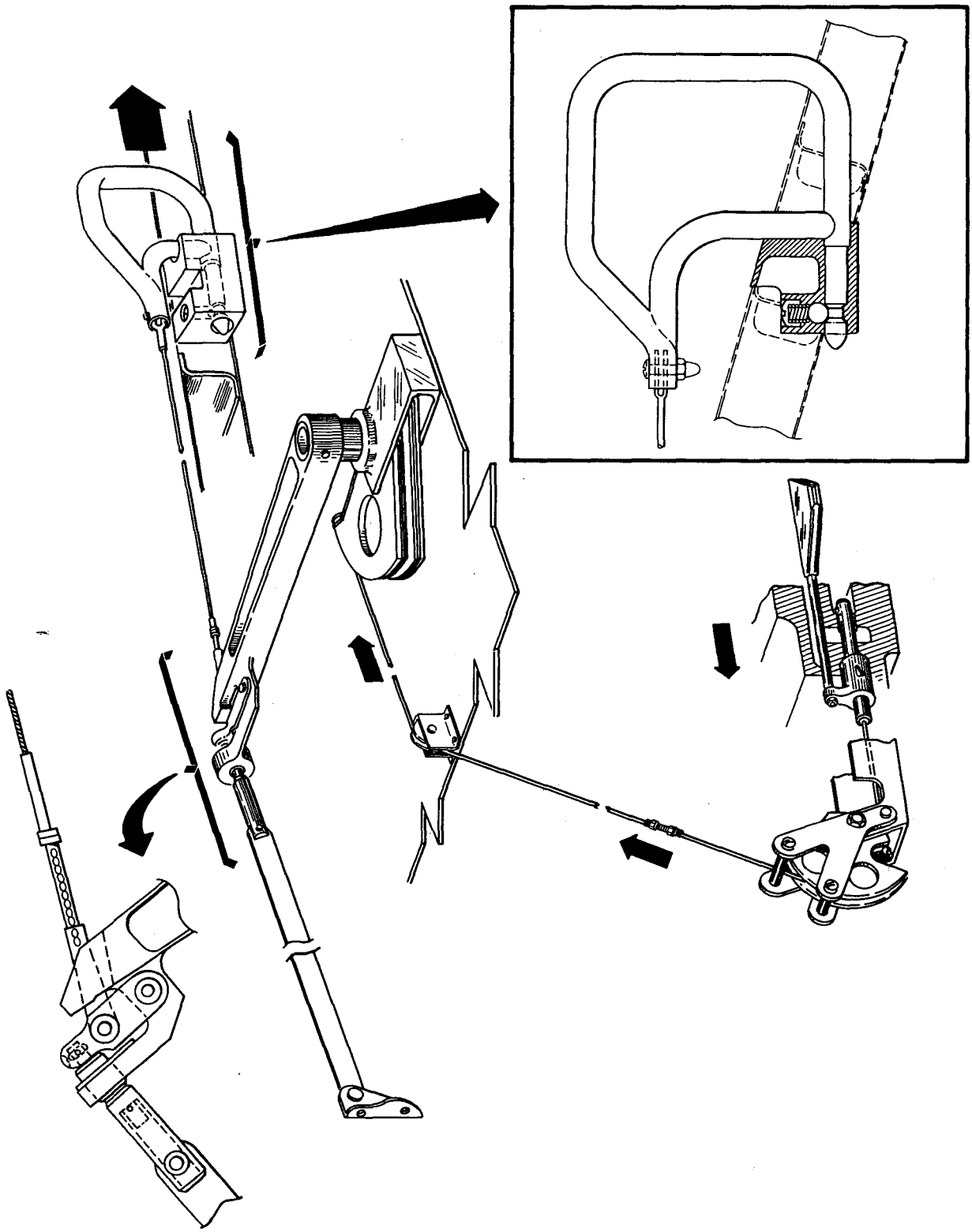
**BACK-UP DEPLOYMENT**

T30005-221



MESA DEPLOYED

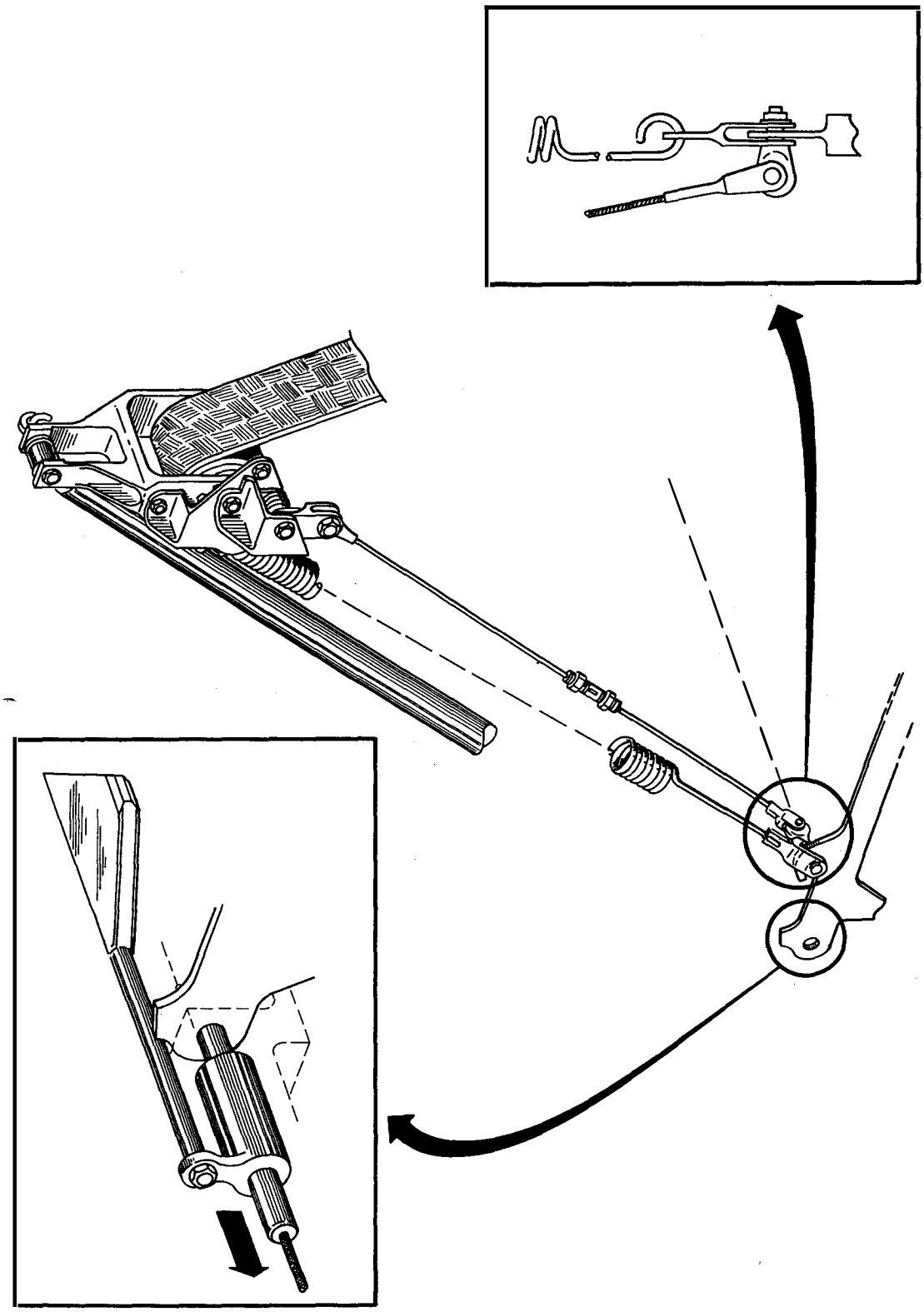
T30005-223.1



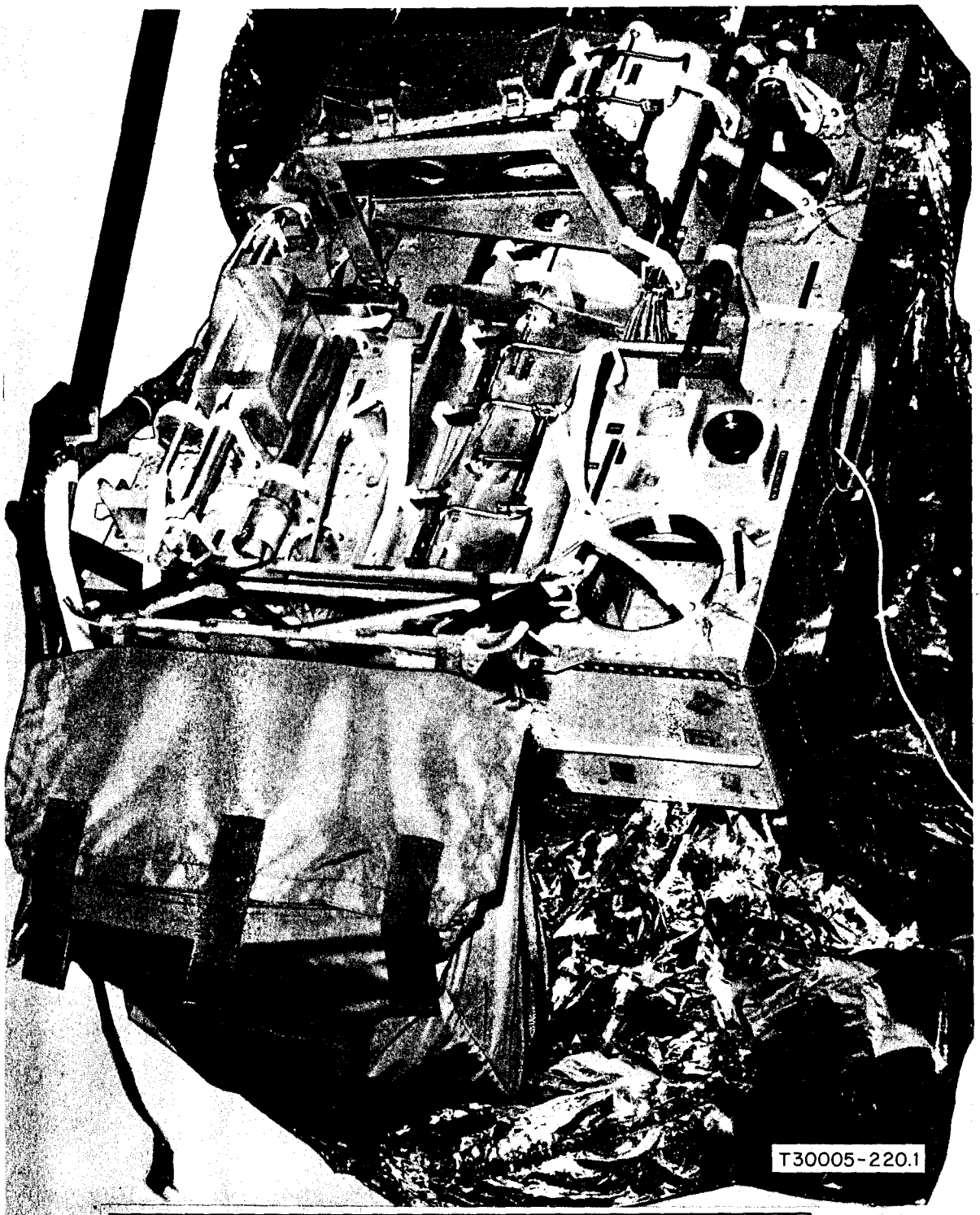
T30005-227

MESA DEPLOYMENT MECHANISM

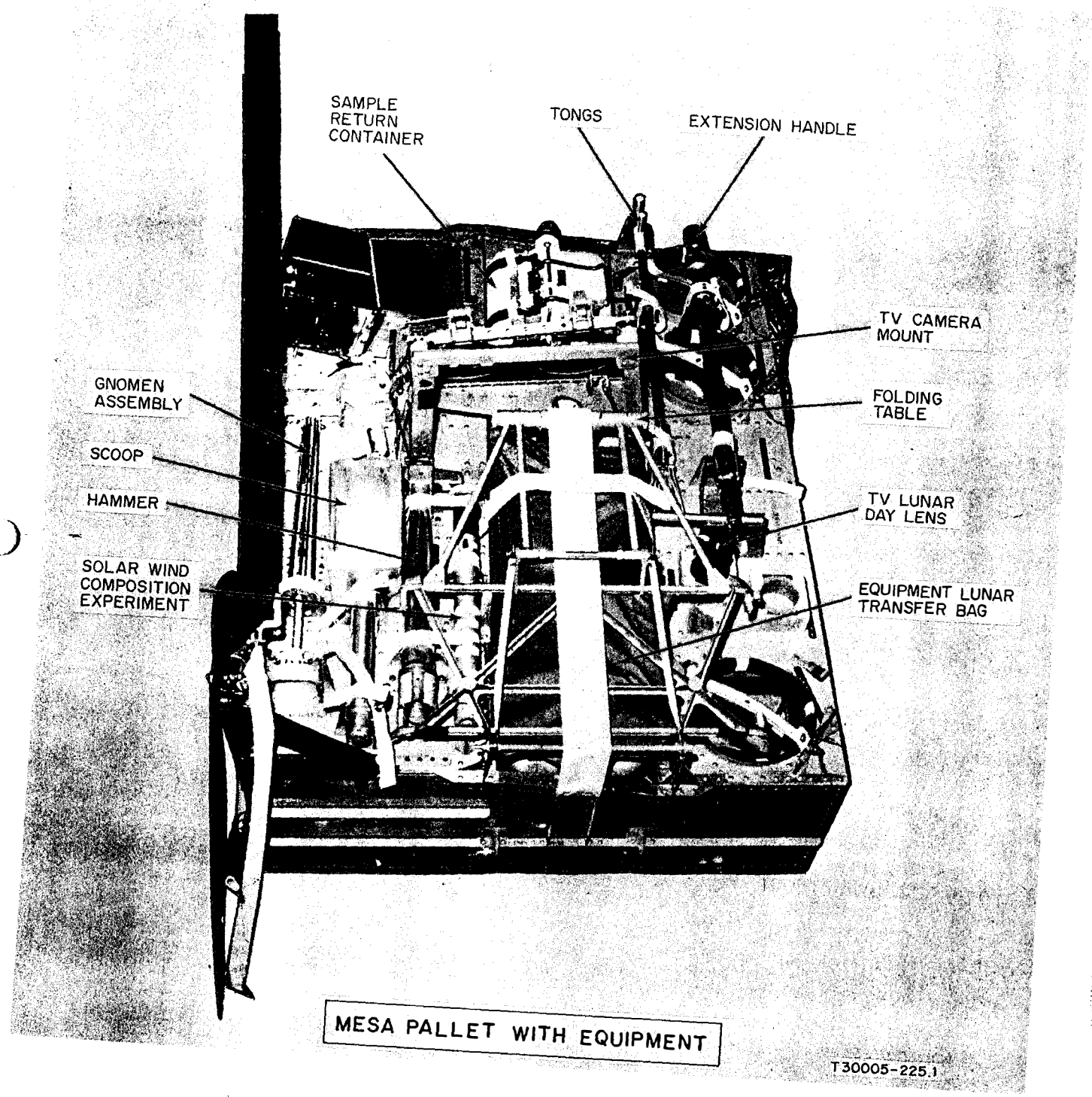
T30005-226



MESA DEPLOYMENT RATCHET/SPRING ASSEMBLY



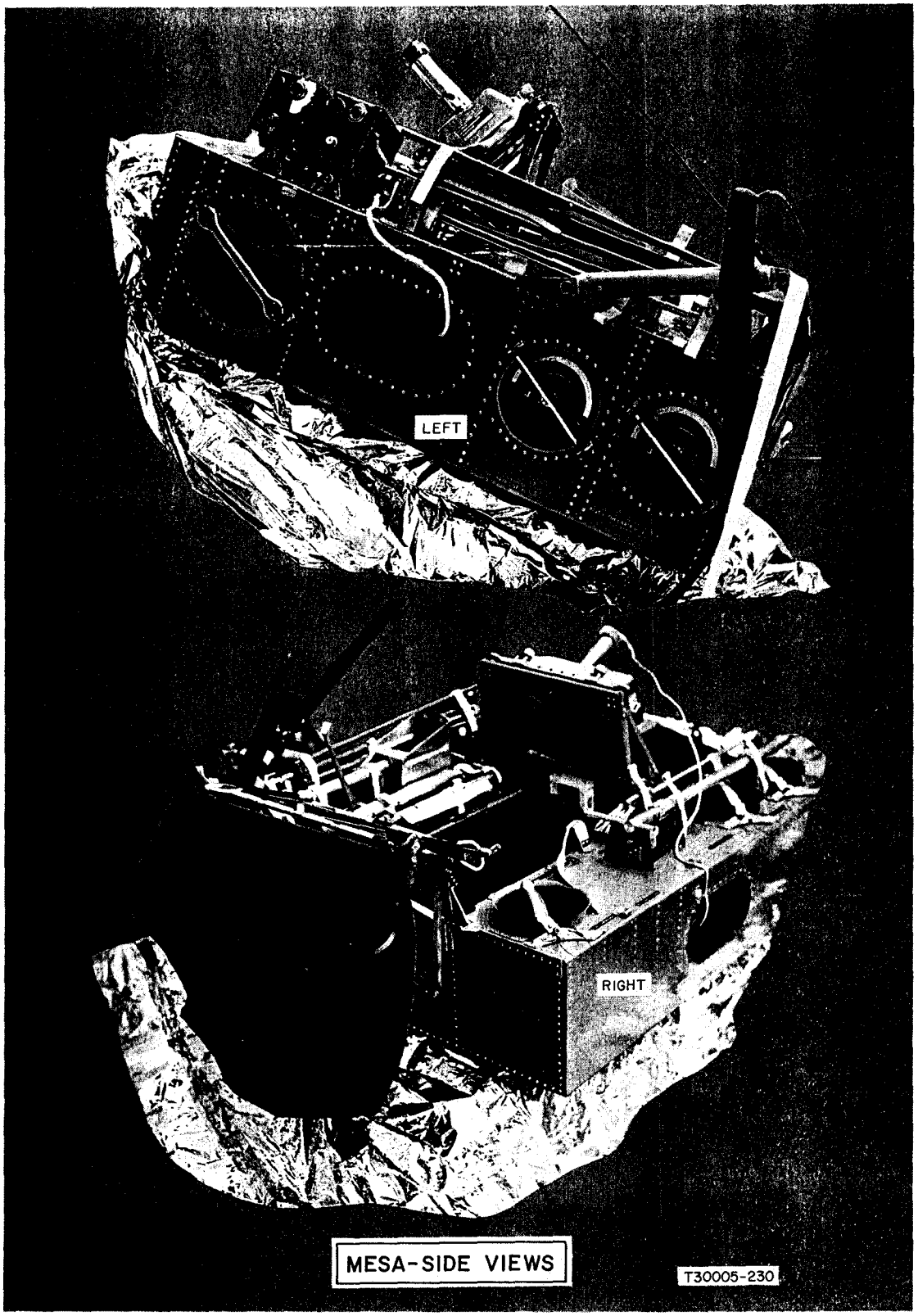

MODULARIZED EQUIPMENT STOWAGE ASSEMBLY
MESA PALLET

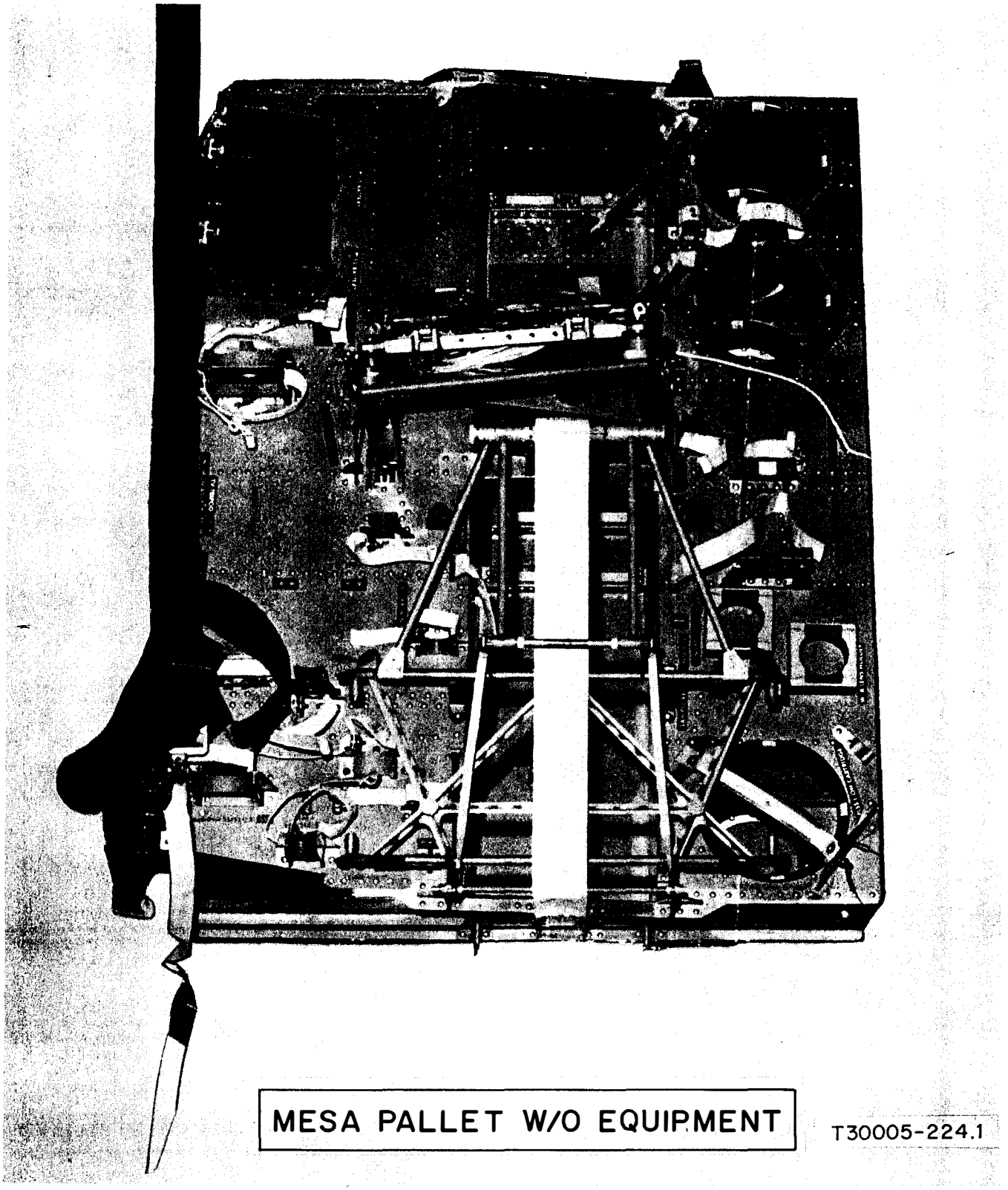



Date: May 1969

# MESA STOWAGE LIST - LM-5

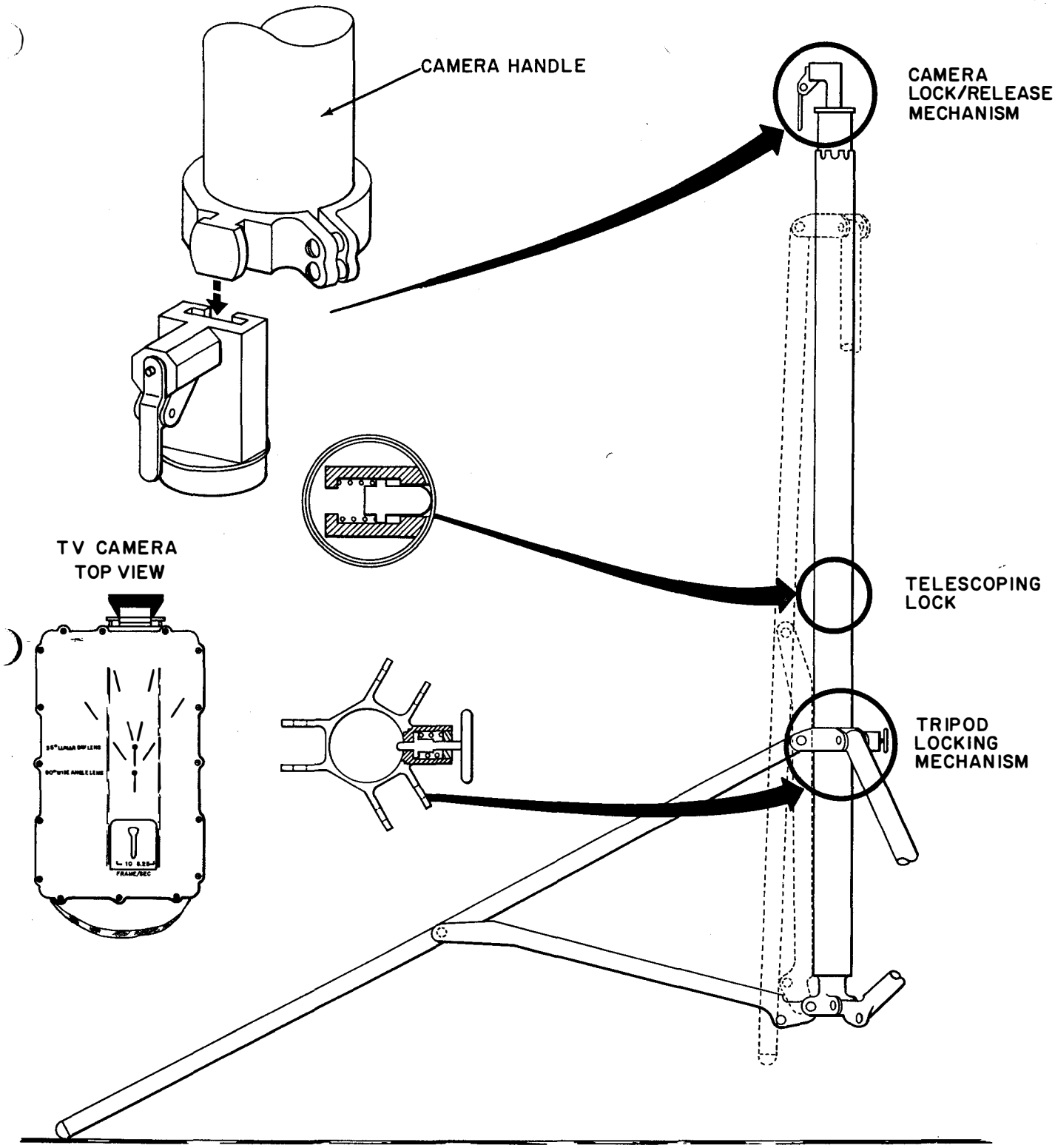
| Item | Weight Lbs. | Qty | Earth Launch | Lunar Launch | LM/CSM Transfer |
|---|---|---|---|---|---|
| Lunar Television Subsystem | | | | | |
| Lunar Television Camera | 7.2 | 1 | * | | |
| TV Wide Angle Lens | .4 | 1 | * | | |
| TV Lunar Day Lens | .4 | 1 | * | | |
| TV Cable Assembly (100 ft) | 2.4 | 1 | * | | |
| TV Tripod Assembly | 1.74 | 1 | * | | |
| Equipment Lunar Transfer Bag | | 1 | * | | |
| S-Band Antenna Cable | 6.0 | 1 | * | | |
| Lunar Sample Scoop | .9 | 1 | * | | |
| Extension Handle | 1.2 | 1 | * | | |
| Tongs | .3 | 1 | * | | |
| Gnomon Assembly | .5 | 1 | * | | |
| Hammer | 1.8 | 1 | * | | |
| Solar Wind Composition Exp | 1.0 | 1 | * | *SRC #2 | *SRC #2 |
| Close Up Stereo Camera | | 1 | * | *Film | *Film |
| Sample Return Container #1 | 15.6 | 1 | * | * | * |
| Scale | | 1 | | | |
| Solar Wind Composition Bag | | 1 | | | |
| Sample Bags | | 5 | | | |
| Packing Material | | | | | |
| Bulk Sample Container | | 1 | | | |
| Sample Return Container #2 | 20.9 | 1 | * | * | * |
| Solar Wind Composition Bag | | 1 | | | |
| Core Tubes | | 2 | | | |
| Documentation Sample Collection Bag | | 1 | | | |
| Sample Bags | | 15 | | | |
| Gas Analysis Container | | 1 | | | |
| Lunar Environment Sample Container | | 1 | | | |
| Packing Material | | | | | |


LEFT
RIGHT
MESA-SIDE VIEWS
T30005-230



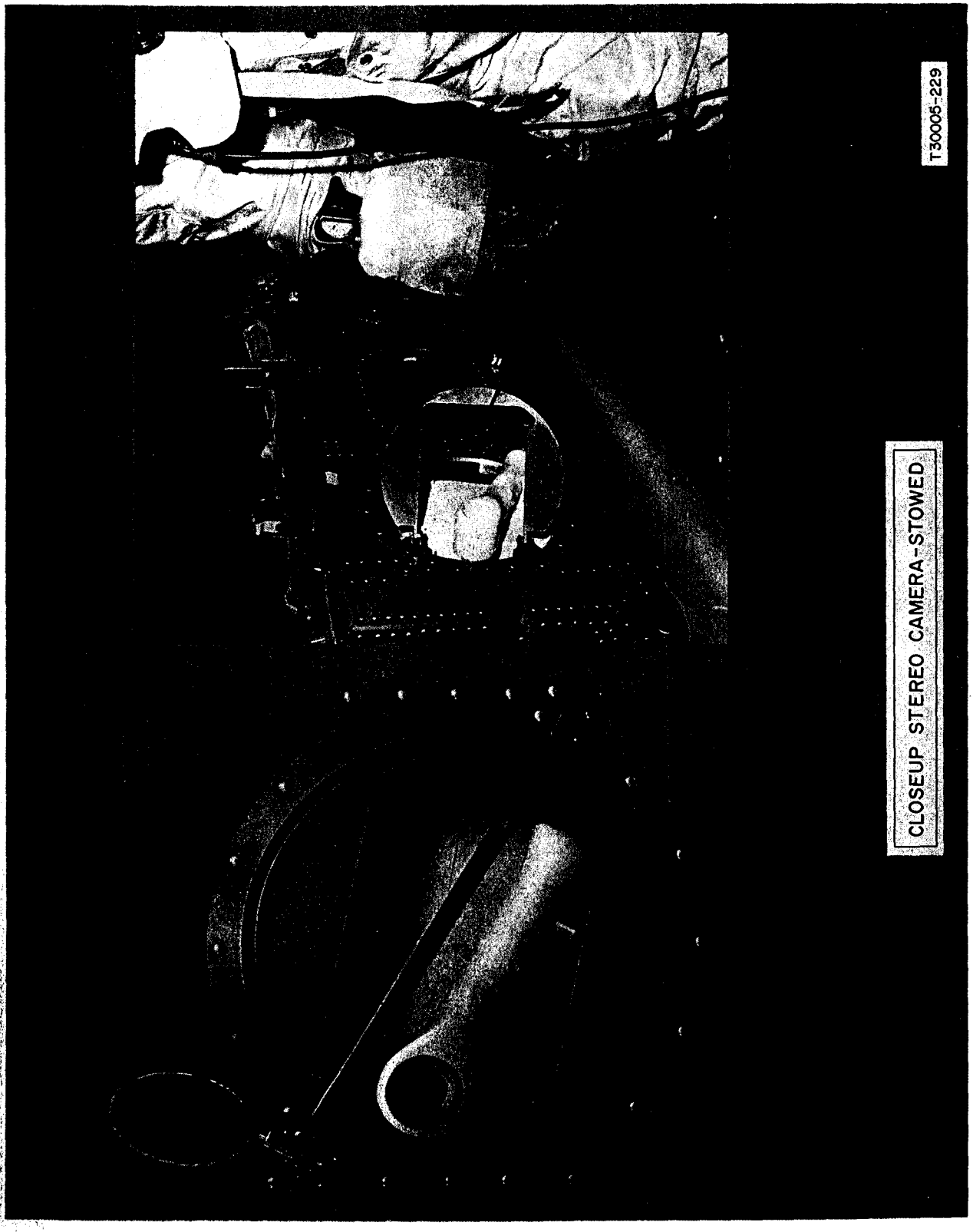
MESA PALLET W/O EQUIPMENT

T30005-224.1



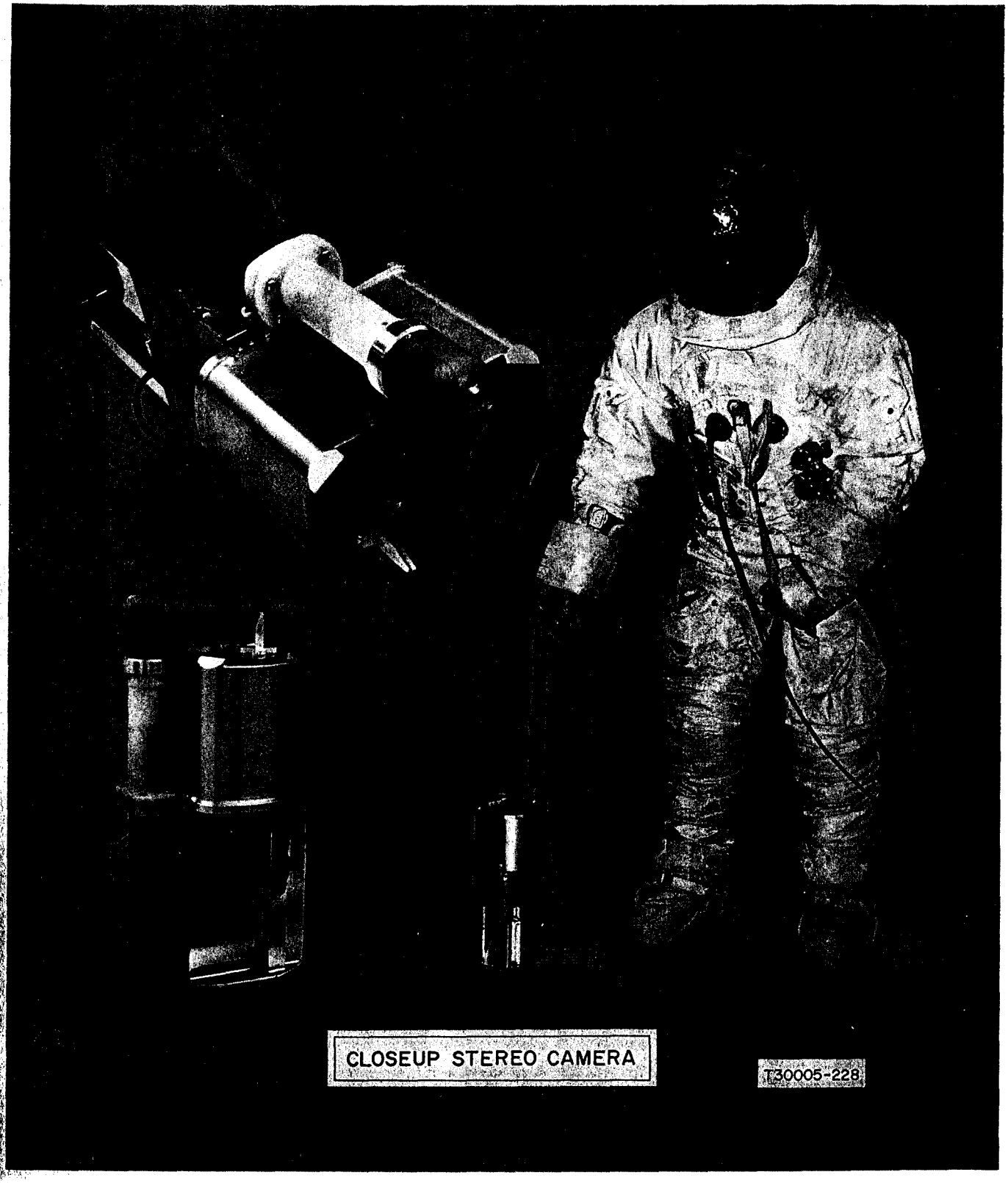

# TV CAMERA AND TRIPOD

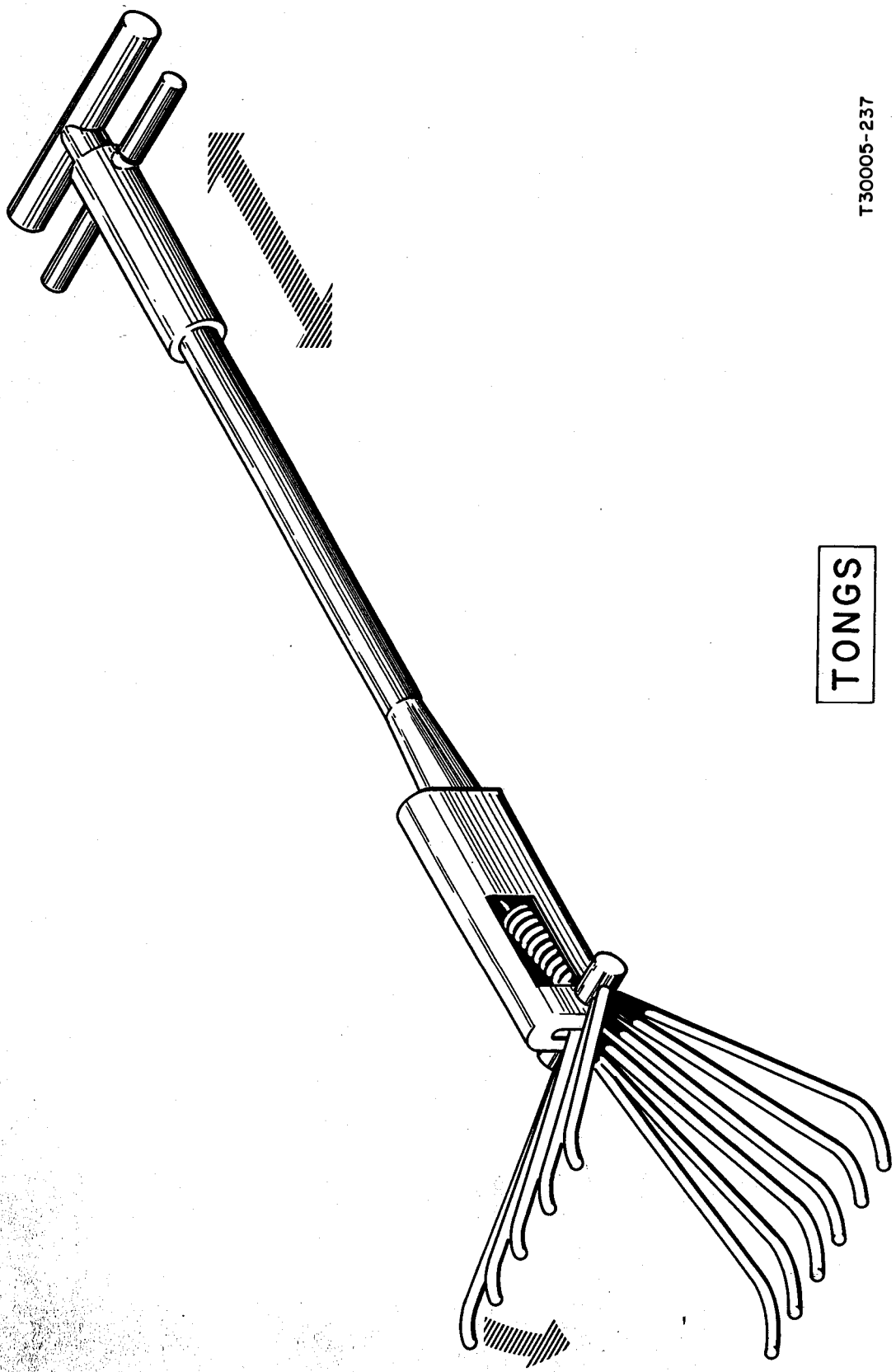
T30915-119.1



T30005-229

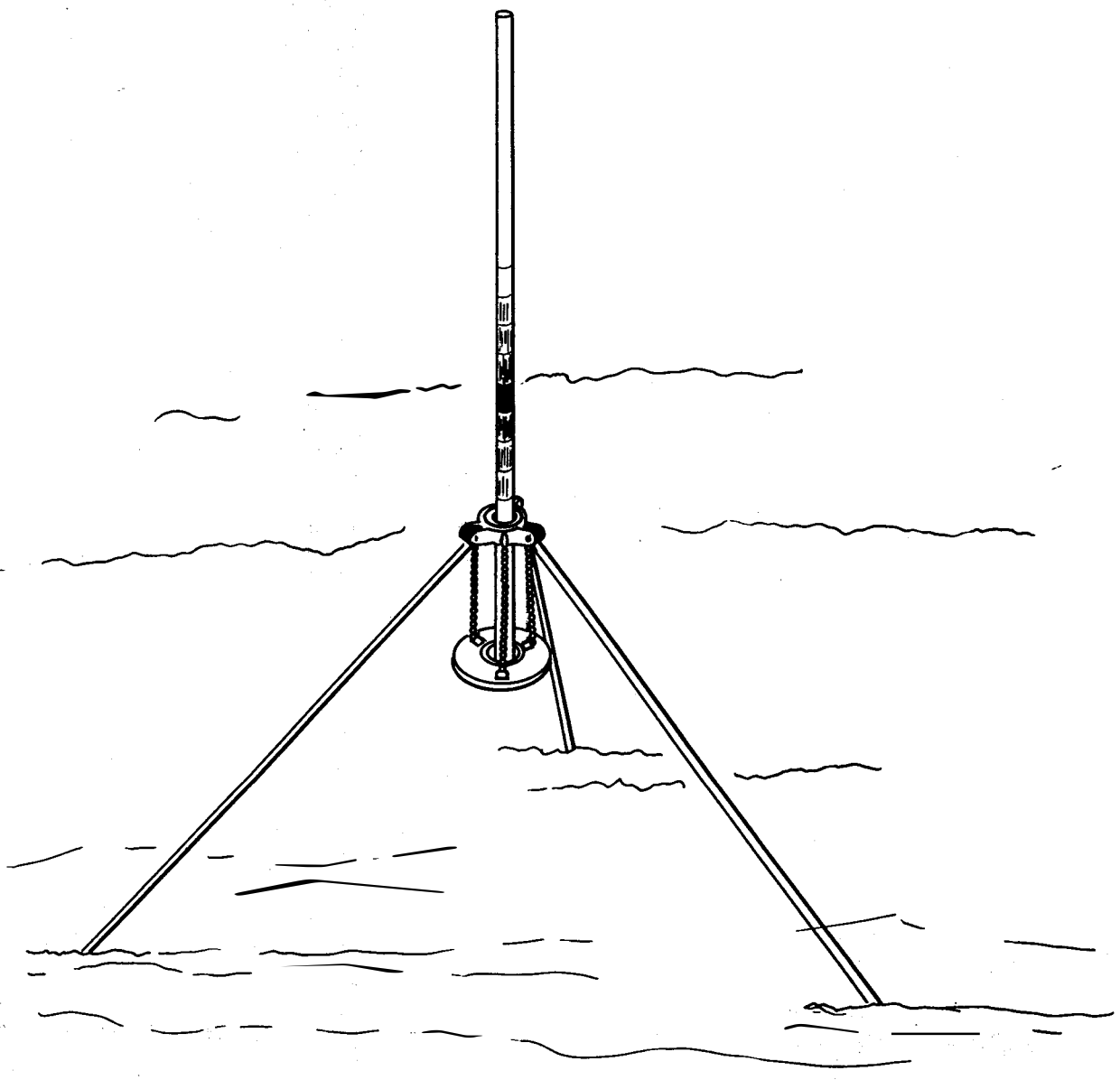
CLOSEUP STEREO CAMERA-STOWED


CLOSEUP STEREO CAMERA
T30005-228



T30005-237

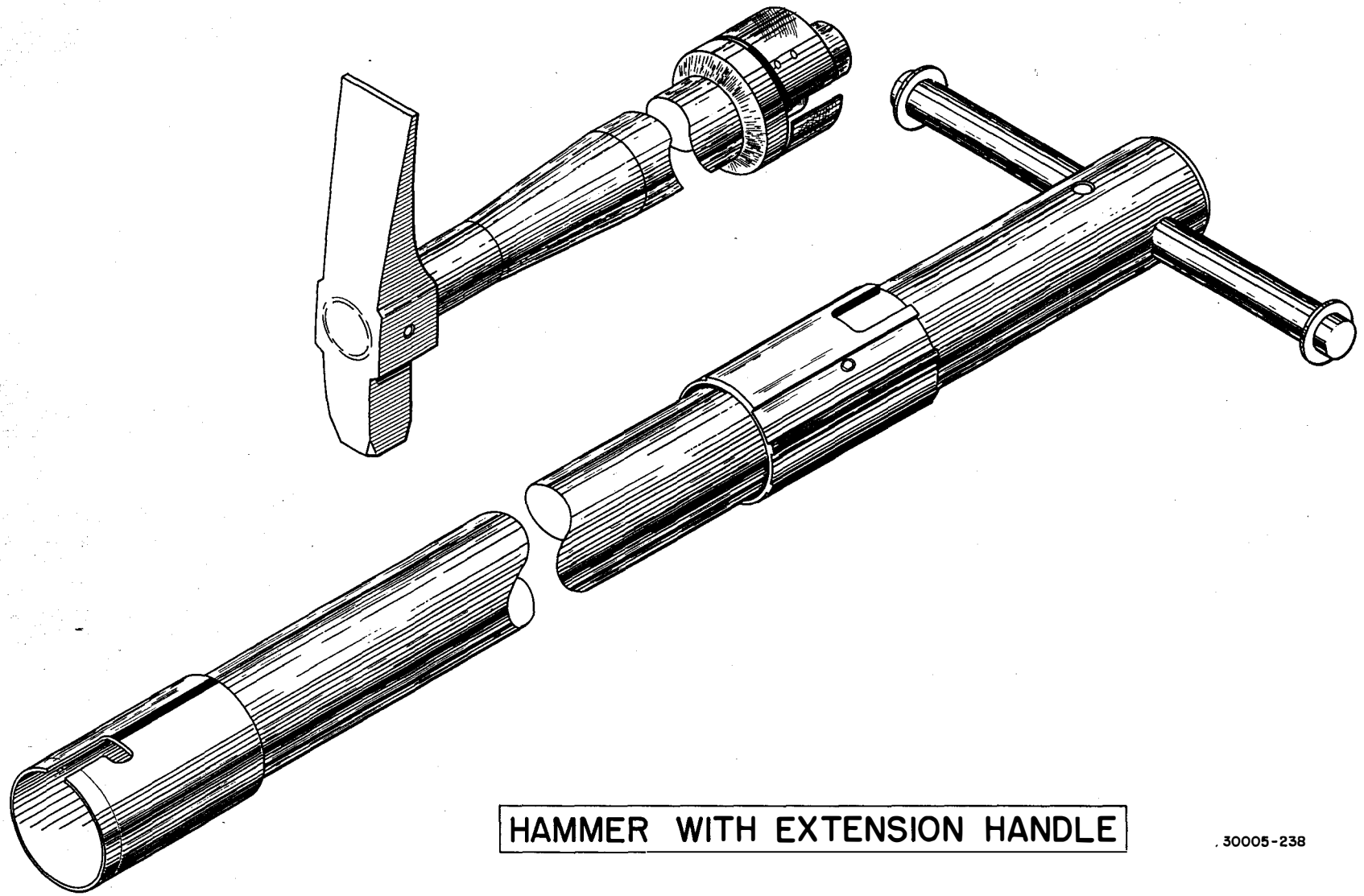
TONGS



GNOMON

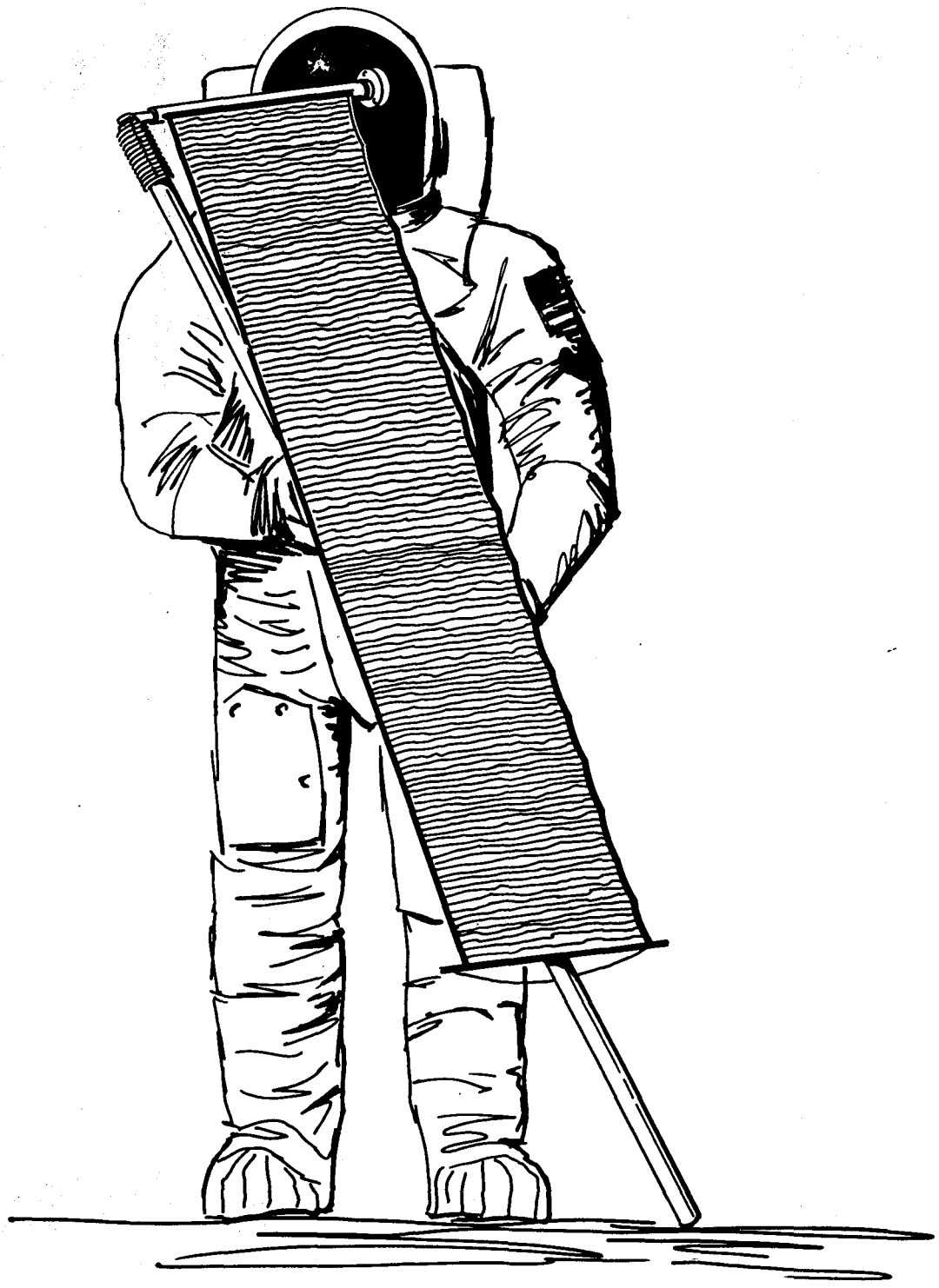
T30005-240

Date: May 1969 FOR TRAINING PURPOSES ONLY



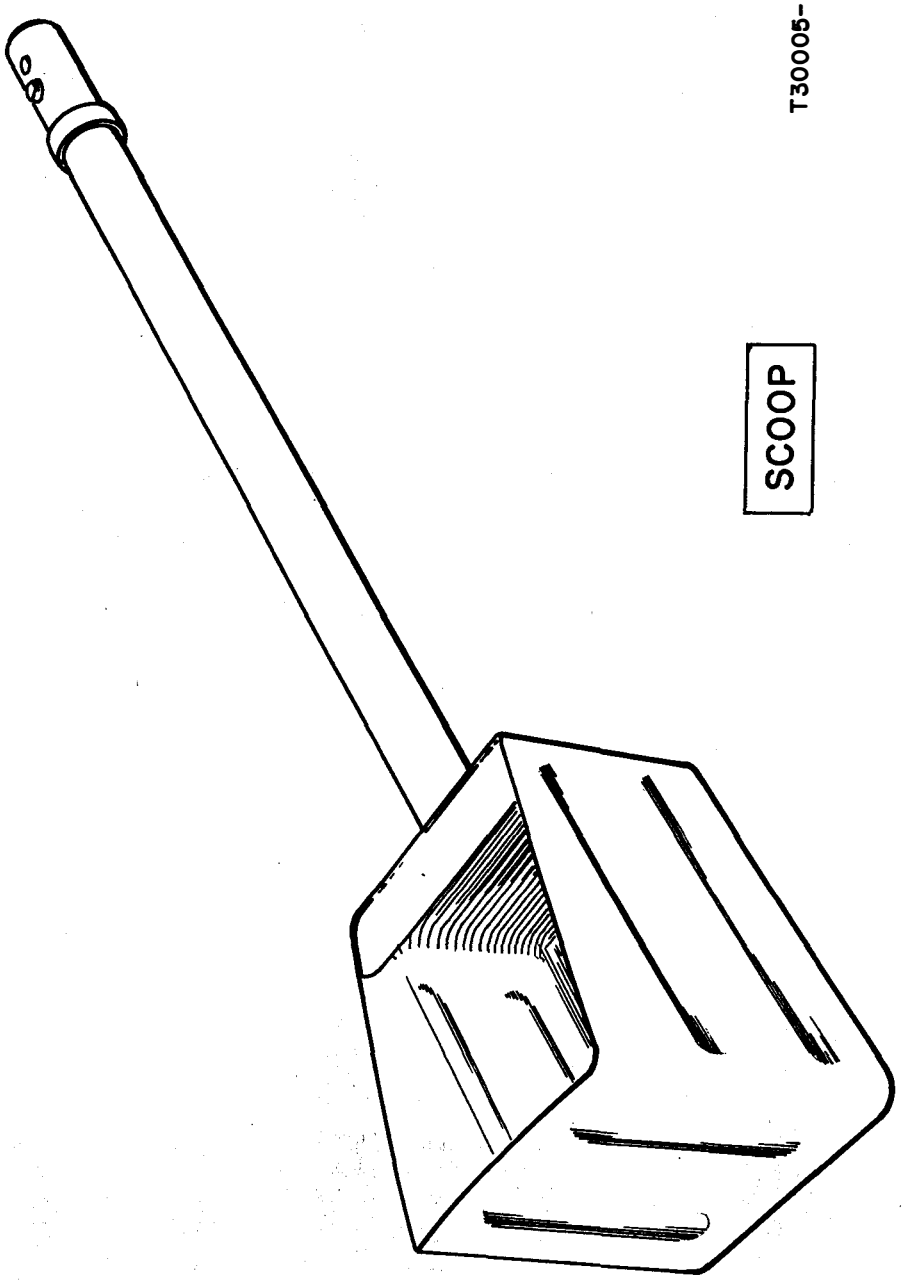
HAMMER WITH EXTENSION HANDLE

30005-238



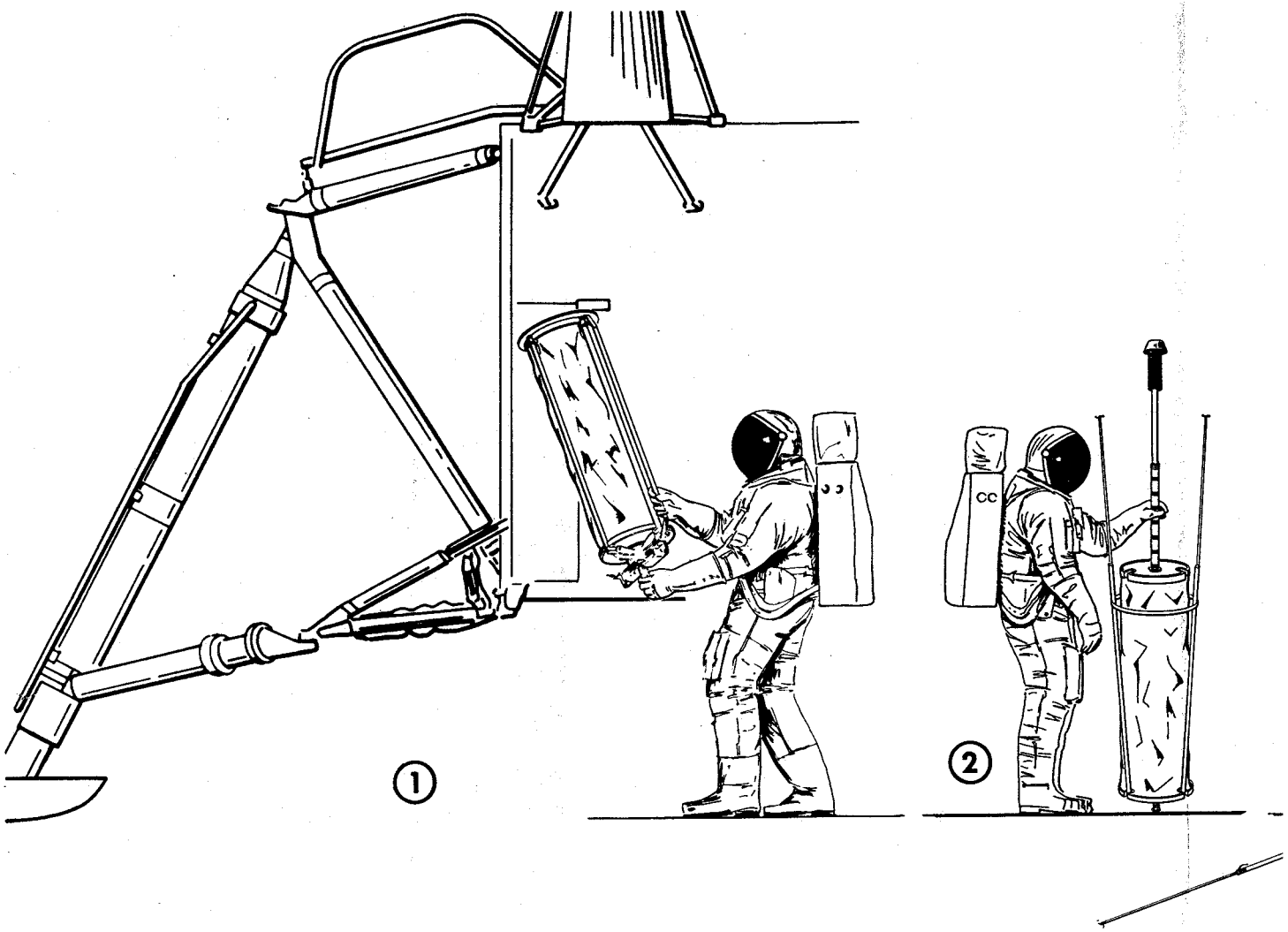
SOLAR WIND COMPOSITION EXPERIMENT

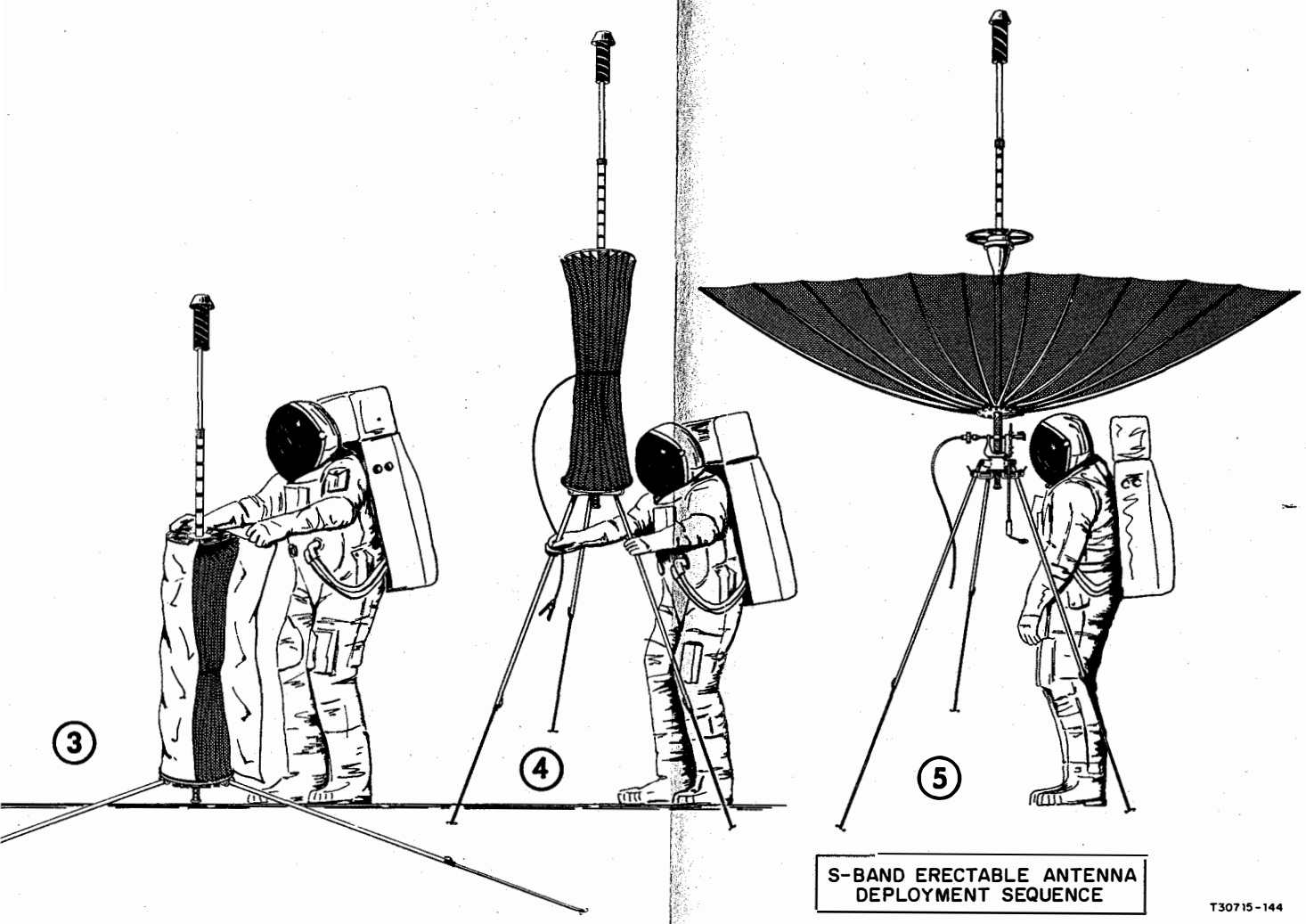
T30005-217



T30005-239

SCOOP


CC
1
2

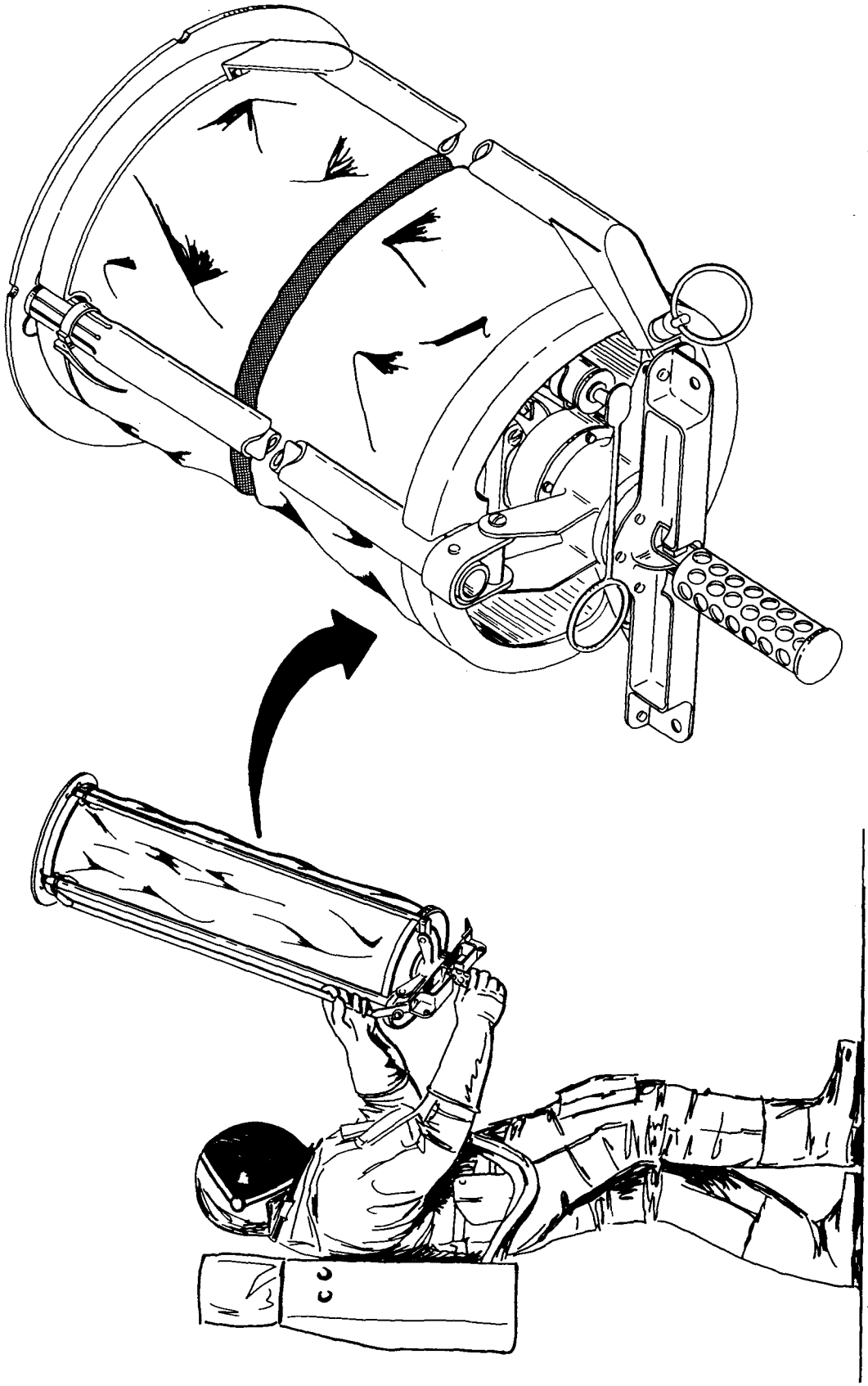



S-BAND ERECTABLE ANTENNA DEPLOYMENT SEQUENCE

T30715-144

T30715-143

# S-BAND ERECTABLE ANTENNA HANDLING

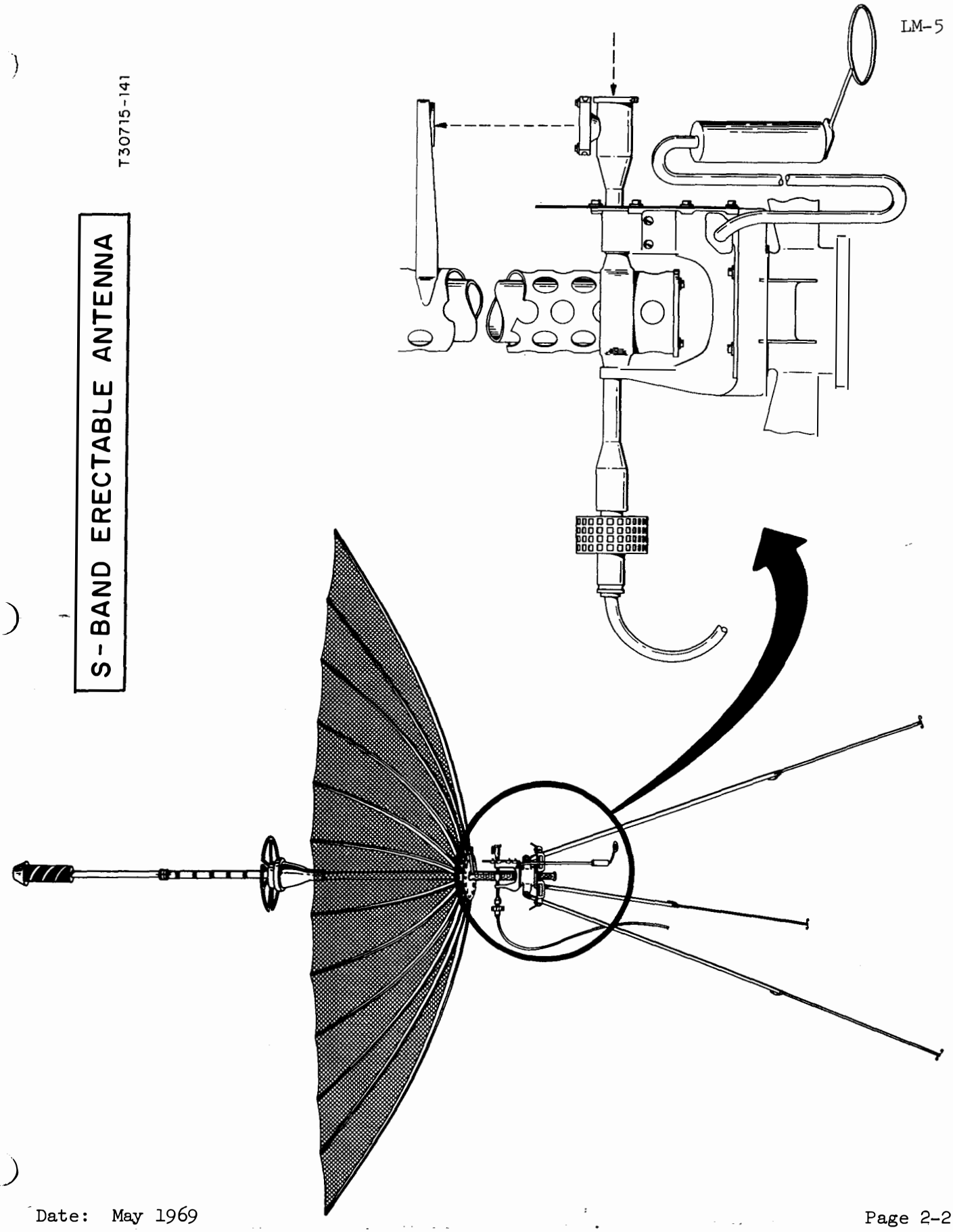


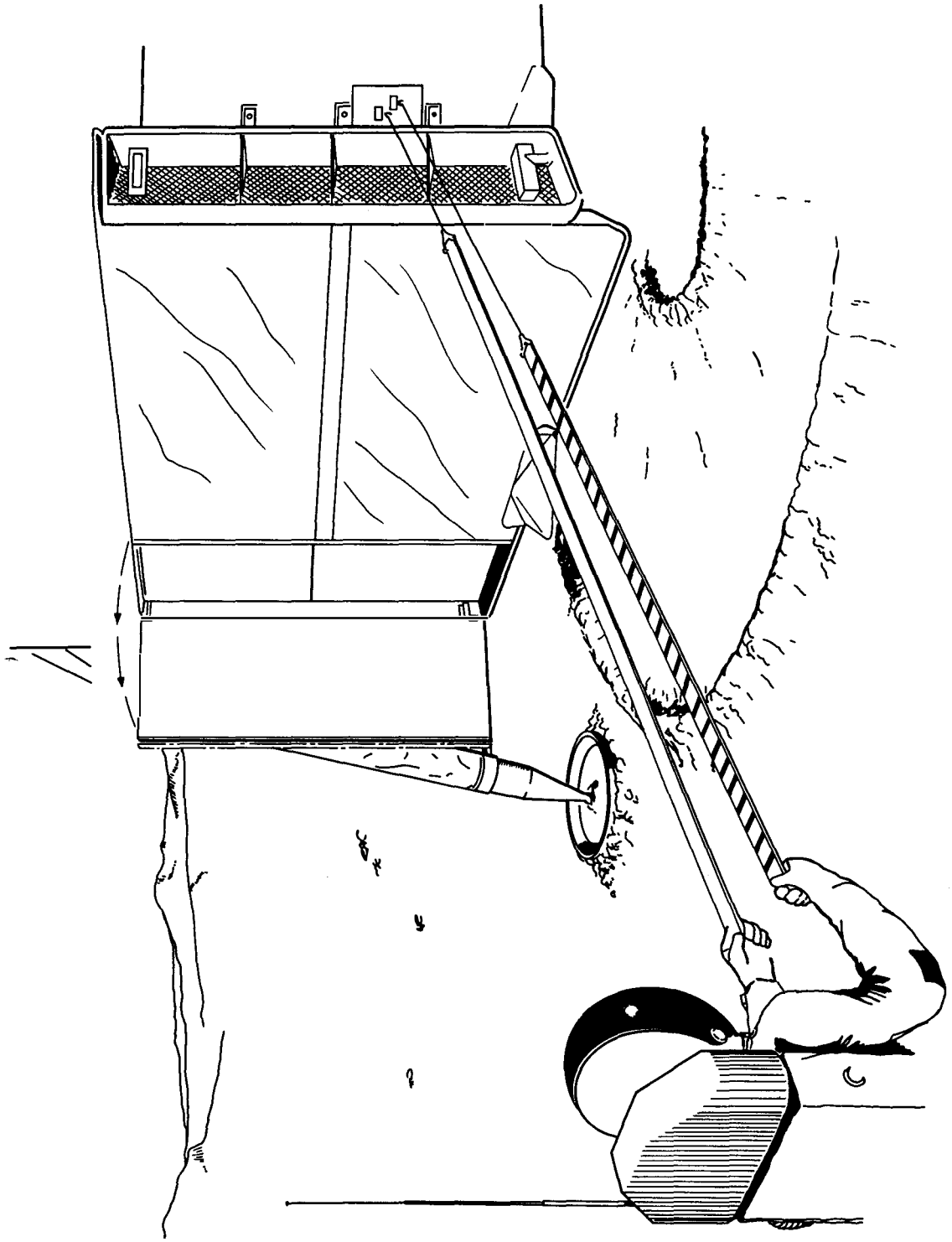


FOR TRAINING PURPOSES ONLY

T30715-141

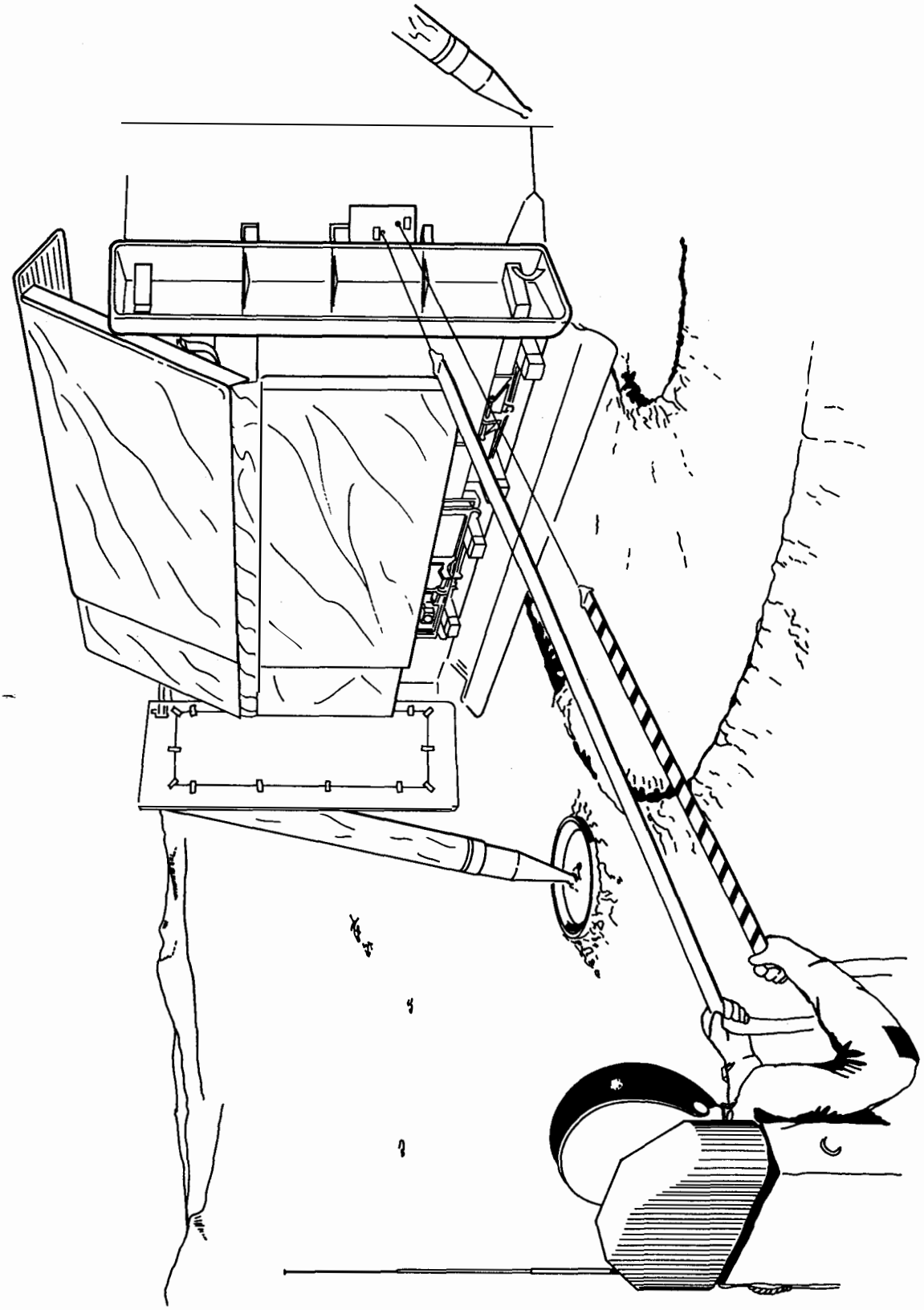
# S-BAND ERECTABLE ANTENNA





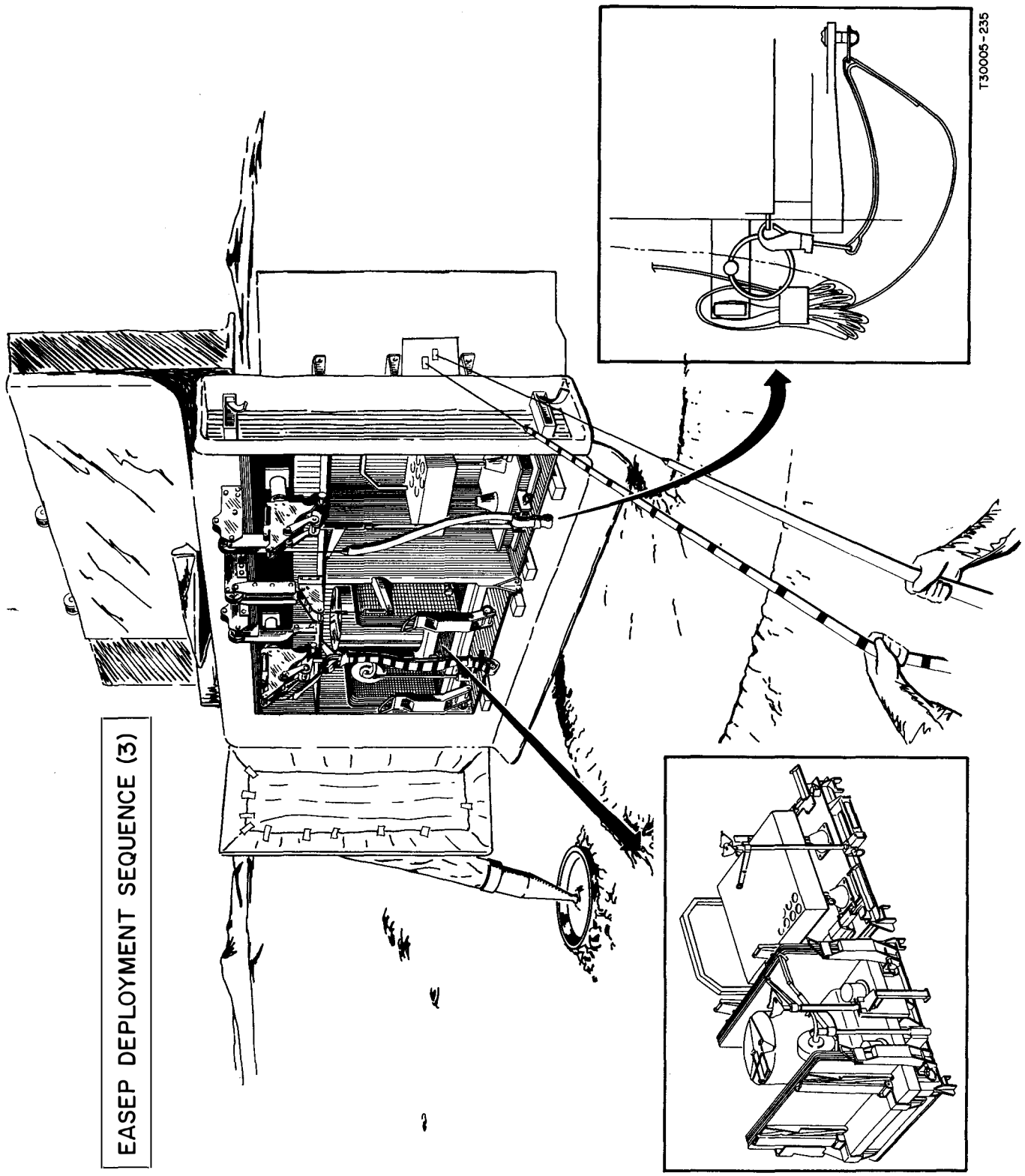
T30005-233

EASEP DEPLOYMENT SEQUENCE (1)



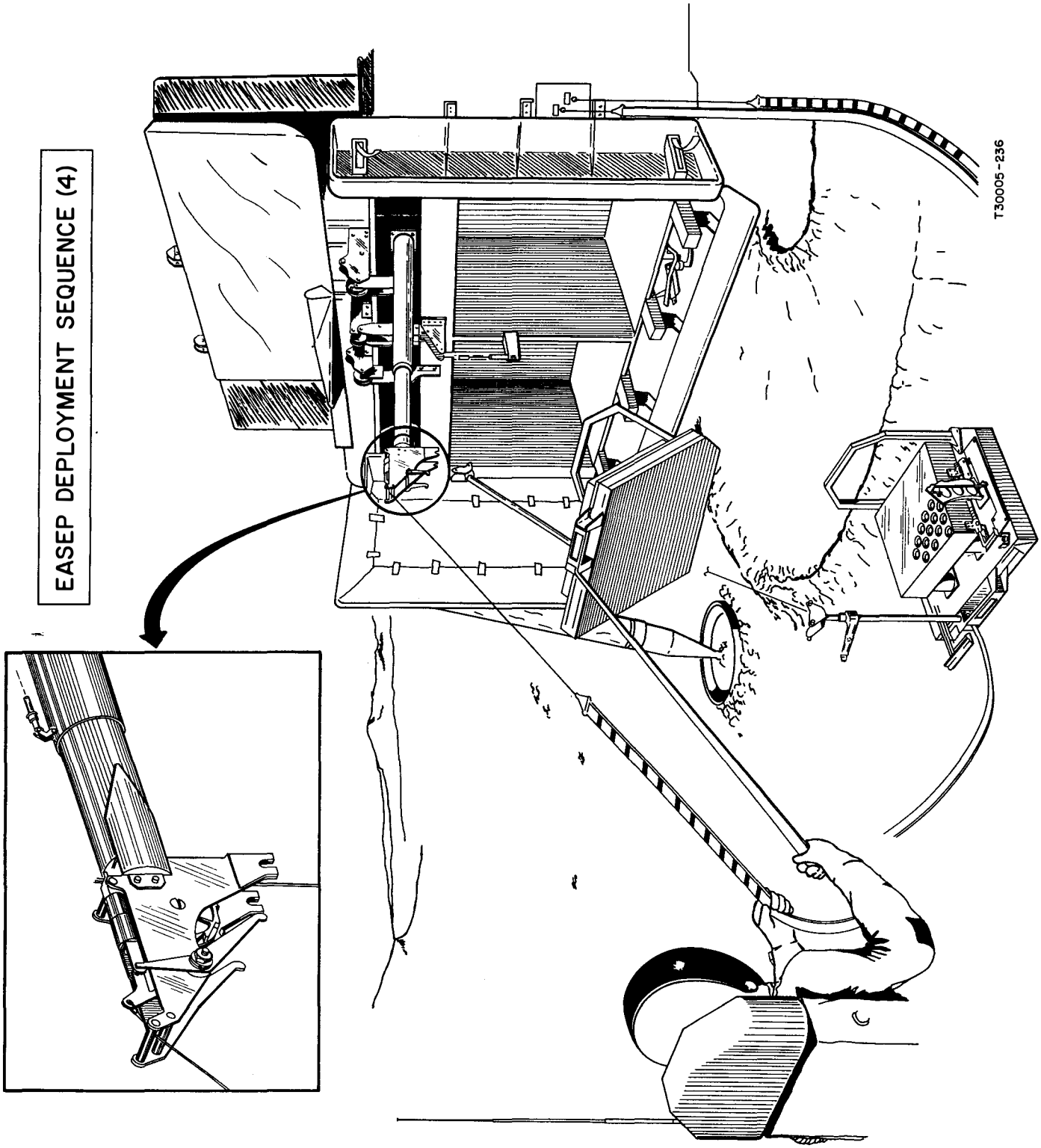
T30005-234

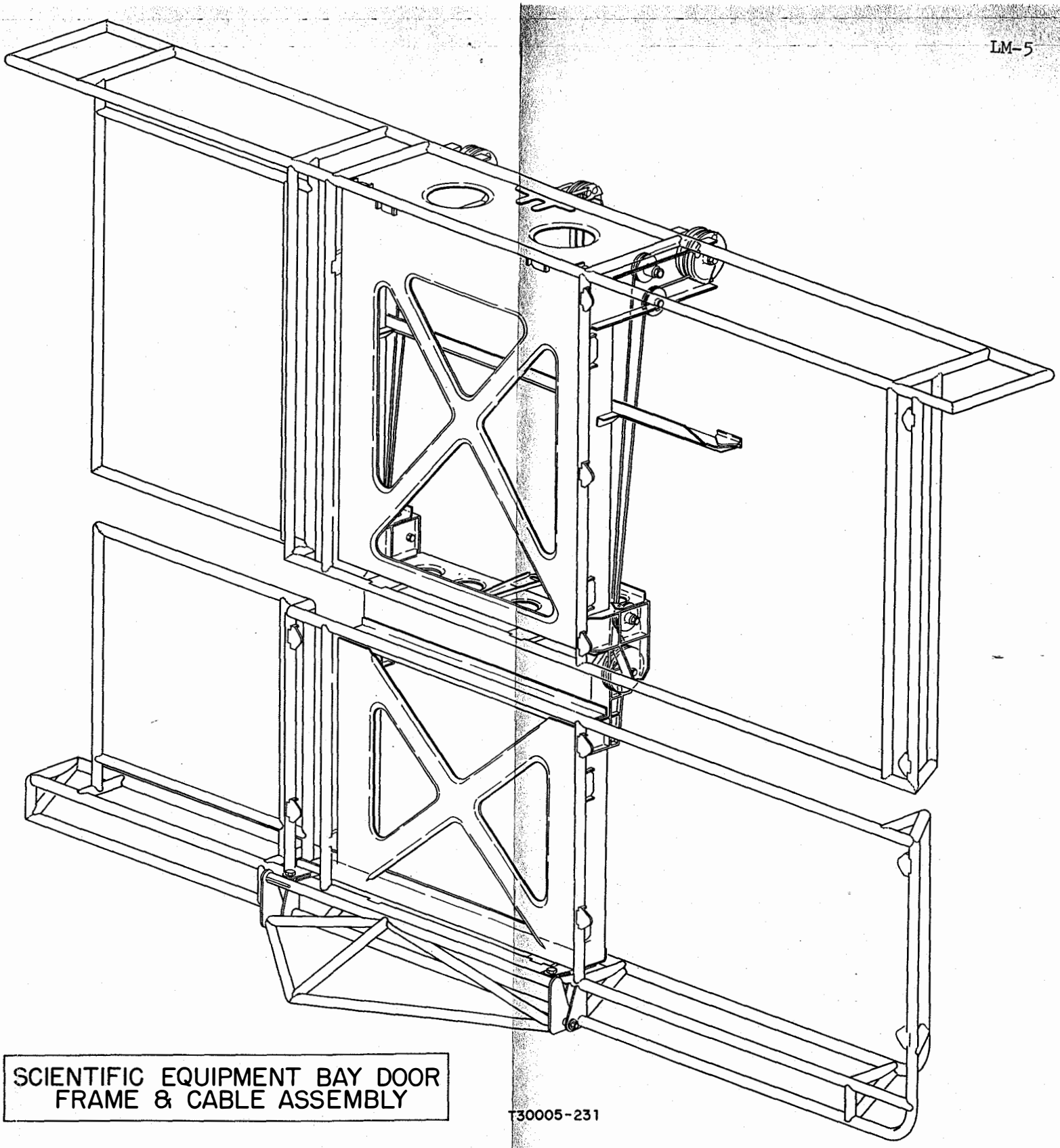
EASEP DEPLOYMENT SEQUENCE (2)




EASEP DEPLOYMENT SEQUENCE (3)

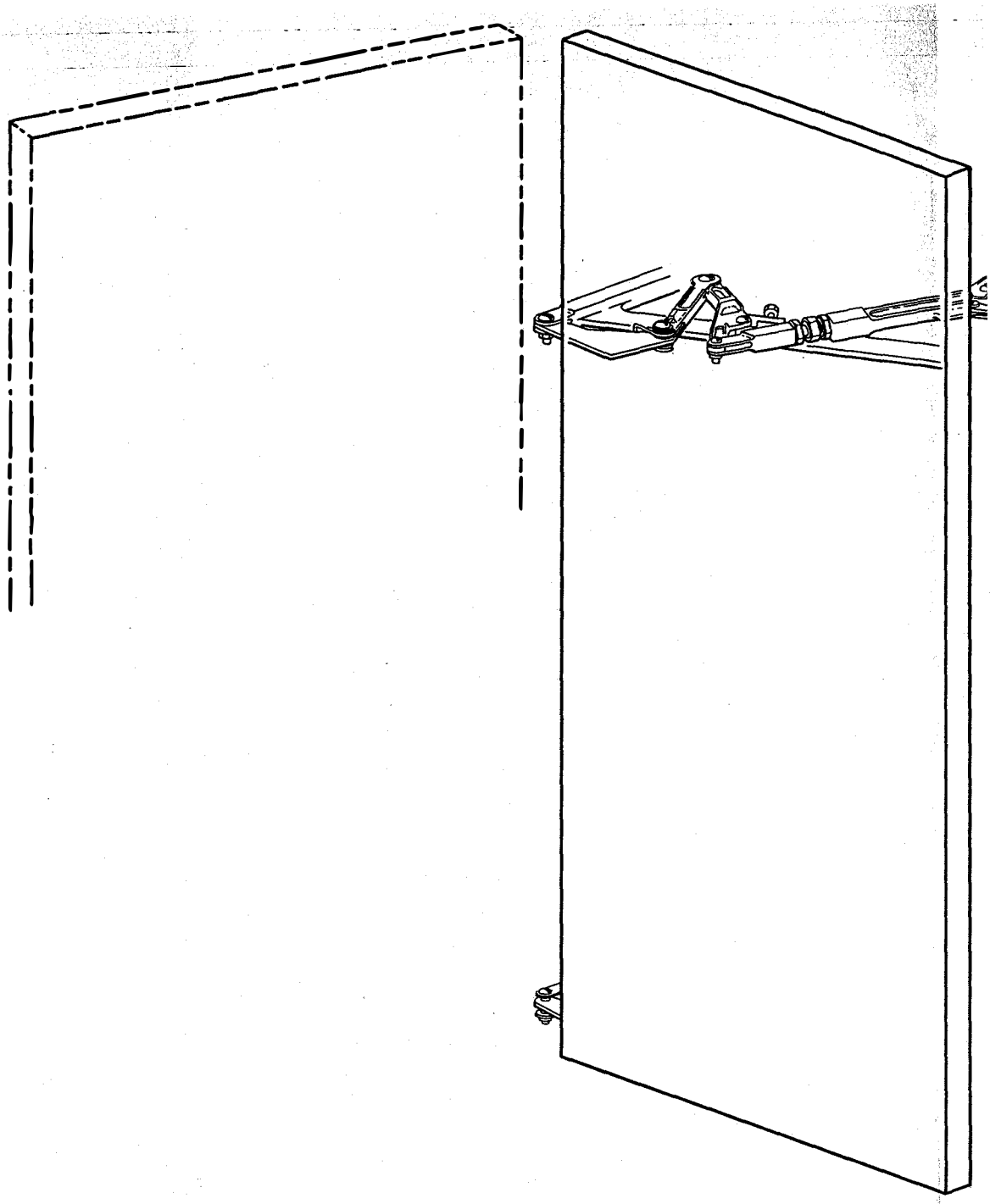
EASEP DEPLOYMENT SEQUENCE (4)

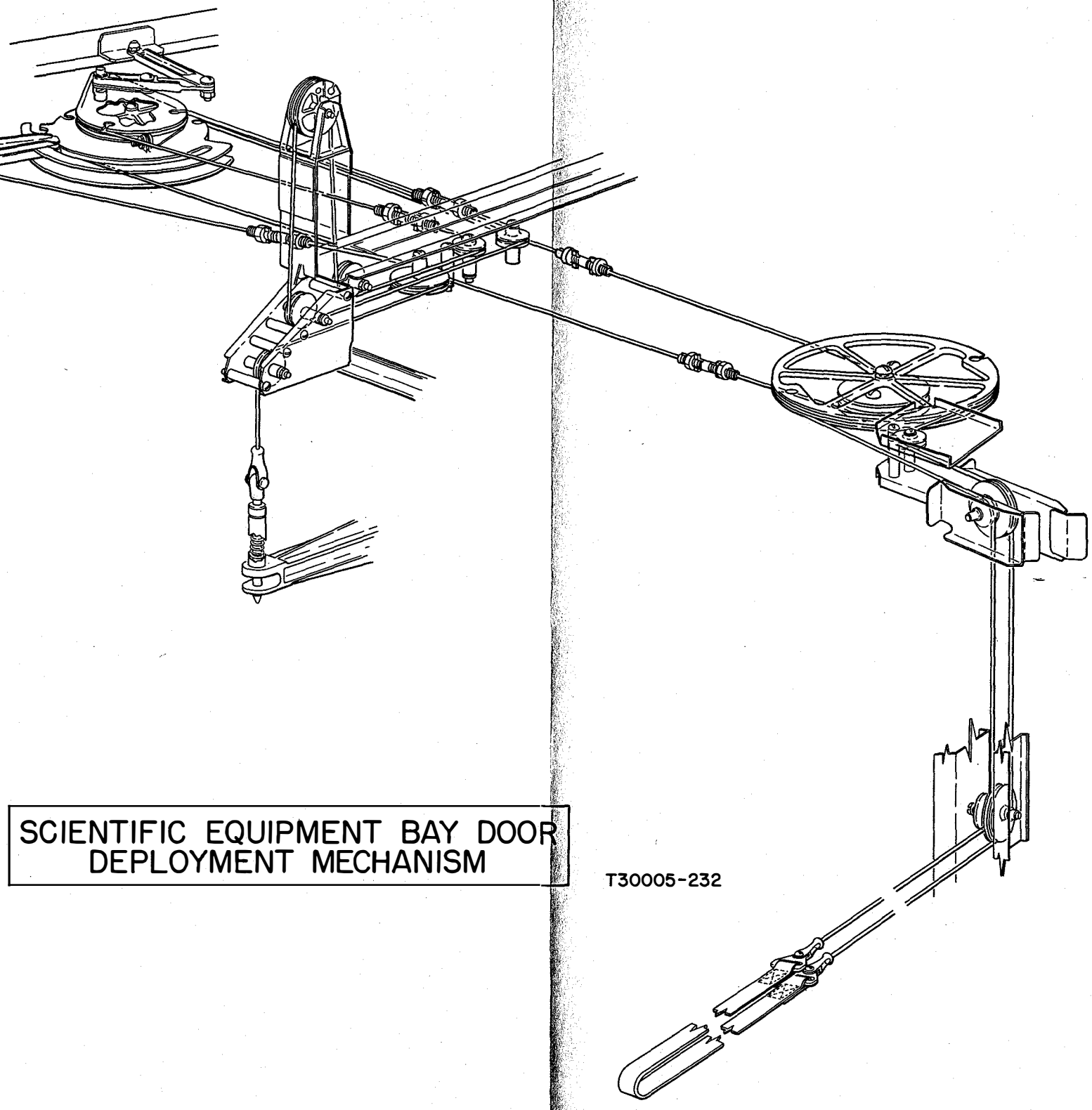




SCIENTIFIC EQUIPMENT BAY DOOR FRAME & CABLE ASSEMBLY

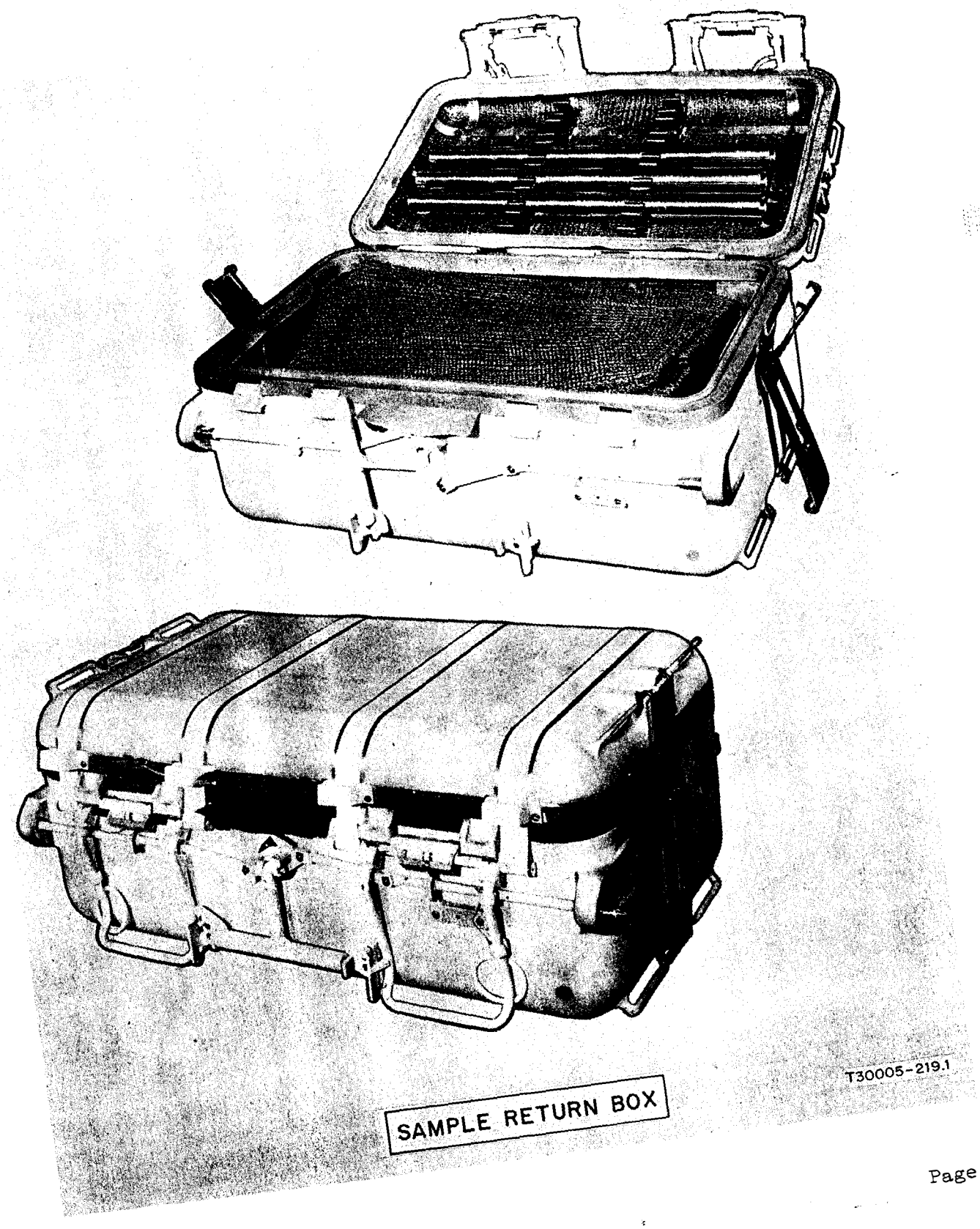
T30005-231



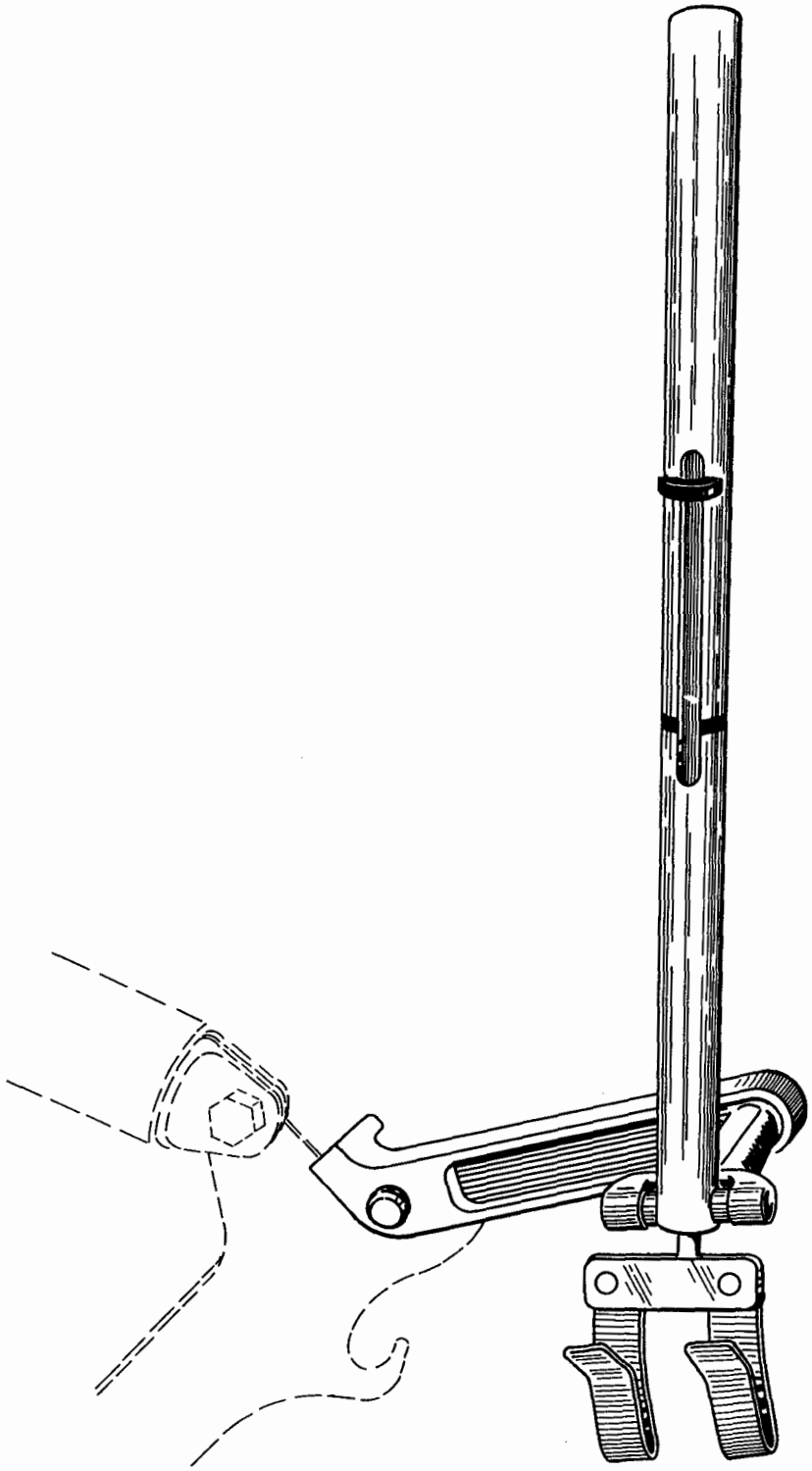


# SCIENTIFIC EQUIPMENT BAY DOOR DEPLOYMENT MECHANISM

T30005-232

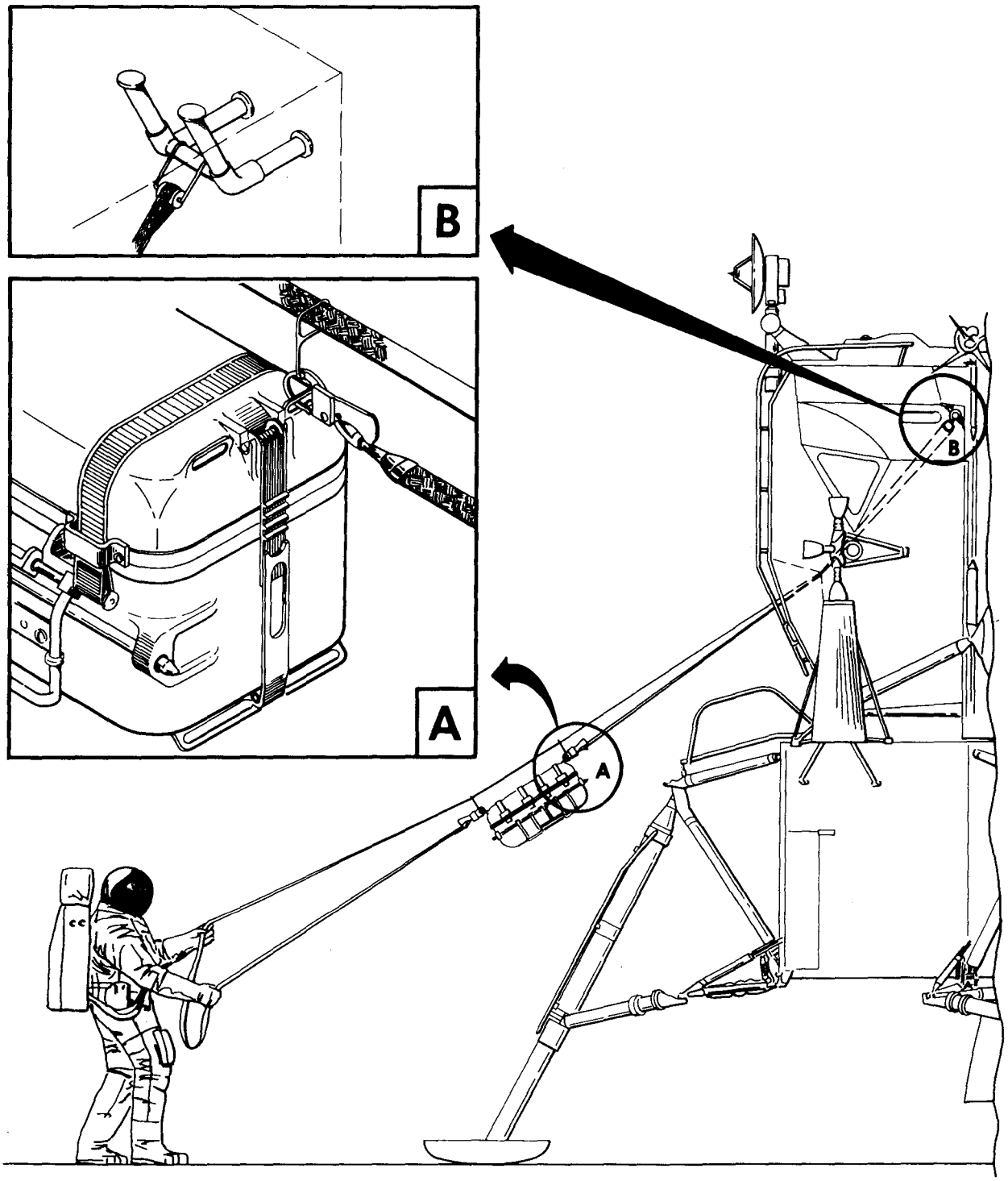



SAMPLE RETURN BOX



SCALE

T30005-241




LUNAR EQUIPMENT CONVEYOR

T30005-218